

Algebraic Topology

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May 8, 2026

Abstract

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Let $\text{Gr}(k, n)$ denote the Grassmannian of k -planes in V , a real or complex vector space of dimension n . We will not distinguish since the statements we make will be true for both cases. For the real Grassmannian, we take homology with $\mathbb{Z}/2$ coefficients, and for the complex Grassmannian, we take homology with $\mathbb{Z}$ coefficients.

Theorem 1.1. For a CW complex X , $\mathbb{R}, \mathbb{C}, \mathbb{H}$ -vector bundles over X up to isomorphism are classified by homotopy classes of maps to the infinite Grassmannian $Gr_k(\mathbb{C}^\infty) = \lim_{n \rightarrow \infty} Gr_k(\mathbb{C}^n)$.

Remark 1.2. $\text{Sp}(n) = U(2n) \cap GL(n, \mathbb{H})$ is the compact symplectic group acts on $\mathbb{H}^n$ by left multiplication and commutes with the right $\mathbb{H}$ -action which is the structure map for $\mathbb{H}$ -vector bundles.

Recall that the Schubert stratification is given by row echelon form for matrices. Let $T \subset V$ be a subspace of dimension k . Write T as the row space of a $k \times n$ matrix. We can perform row operations to put it in row echelon form, and this depends only on T . In particular, T has a canonical basis but it does not depend continuously on T . It may jump whenever the pivots move. The Schubert stratification is given by the locus where the pivots are fixed. Therefore, it is enough to give an increasing sequence σ_p of pivot positions or a sequence s of 0's and 1's of length n with exactly k 1's.

Example 1.3. Let us row reduce from right to left. For example in $\text{Gr}(2, 4)$

$$\begin{bmatrix} * & 0 & * & 1 \\ * & 1 & 0 & 0 \end{bmatrix}$$

corresponds to the sequence $s = (0, 1, 0, 1)$ and the pivot positions $\sigma_1 = 2, \sigma_2 = 4$. The dimension is given by

$$\dim S_\sigma = \sum_{p=1}^k (\sigma_p - p) = (2 - 1) + (4 - 2) = 3 = \sum_{1s \in s} \text{total number of 0s to the left} = 1 + 2$$

There are two inclusions called i, j of $\text{Gr}(k, n)$ into $\text{Gr}(k, n + 1)$ and $\text{Gr}(k + 1, n + 1)$ given by $T \mapsto T \oplus e_{n+1}$ and $T \mapsto e_0 \oplus T$ respectively.

1.1 Some generalities

These cells define a CW decomposition. They give a basis of $\mathbb{Z}$ homology for $\mathbb{C}, \mathbb{H}$ or $\mathbb{Z}/2$ homology for $\mathbb{R}$. It is obvious for $\mathbb{C}, \mathbb{H}$ because there are no attaching maps, but for $\mathbb{R}$ you have to argue that the attaching maps have even degree. The closure of S_σ is the union of cells S_τ if $\tau_p \leq \sigma_p$ for all p meaning no pivot of σ is to the left of the corresponding pivot of τ . Poincare duality is implemented by reverseing the sequence of 0s and 1s, reverse the order of the basis and check the intersection of the cells. Most $\overline{S_\sigma}$ are singular but the singularities have codimension at least 2 and thus define fundamental classes and can use the intersection pairing to check Poincare duality. In particular they are normal varieties.

1.2 The inclusions i, j

What are the smallest cells i, j are missing? For i we are missing

$$[F_k | M | e_0]$$

where F_k is the antidiagonal identity, M is stars in the top row and 0s elsewhere, and e_0 is $1, 0, \dots, 0$ in the last column. This corresponds to the sequence $s = 11 \dots 10 \dots 01$ and is of dimension $n - k$.

For j we are missing

$$[* | F_{k+1} | 0]$$

corresponds to the sequence $01 \dots 10 \dots 0$ and is of dimension $k + 1$.

Corollary 1.4 (Approximation theorem). If $\dim X < n - k$ or $\dim X < 2(n - k)$ in the complex case, then a map $X \rightarrow \text{Gr}_{\mathbb{R}}(k, n + 1)$ or $X \rightarrow \text{Gr}_{\mathbb{C}}(k, n + 1)$ can be homotoped to a map $X \rightarrow \text{Gr}(k, n)$. In

particular $\mathrm{Gr}(k, n+1)/\mathrm{Gr}(k, n)$ is $(n-k-1)$ -connected or $(2(n-k)-1)$ -connected in the complex case.

If $\dim X < k+1$ or $\dim X < 2(k+1)$ in the complex case, then a map $X \rightarrow \mathrm{Gr}_{\mathbb{R}}(k+1, n+1)$ or $X \rightarrow \mathrm{Gr}_{\mathbb{C}}(k+1, n+1)$ can be homotoped to a map $X \rightarrow \mathrm{Gr}(k, n)$. In particular $\mathrm{Gr}(k+1, n+1)/\mathrm{Gr}(k, n)$ is k -connected or $2k+1$ -connected in the complex case.

Corollary 1.5 (Stability for classification of vector bundles). If $\dim X < n-k-1$ or $\dim X < 2(n-k)-1$ in the complex case, then $\mathrm{Vect}^k(X) = [X, \mathrm{Gr}(k, n)]$.

Note that we lost one dimension because we need to leave room for the homotopy.

Corollary 1.6. If $\dim X < k+1$ or $\dim X < 2(k+1)$ in the complex case, then any vector bundle of rank $k+1$ has a nonvanishing section.

This also follows from transversality.

1.3 Stiefel manifolds and classifying spaces

Recall that $T^k \rightarrow \mathrm{Gr}(k, n)$ is the tautological subbundle of $\mathbb{R}^n$ or $\mathbb{C}^n$. We have the frame bundle $\mathrm{St}(k, n)$ given by basis of k -planes. Gram Schmidt tells us that orthonormal frames $\hookrightarrow$ frames is a homotopy equivalence so we have $\mathrm{St}(k, n) \rightarrow \mathrm{Gr}(k, n)$ is a principal GL_k -bundle and $\mathrm{St}^{on}(k, n) \rightarrow \mathrm{Gr}(k, n)$ is a principal $O(k)$ -bundle or $U(k)$ -bundle.

What is the codimension of the complement of $\mathrm{St}(k, n)$ in $\mathrm{Mat}_{k,n}$? Well $\mathrm{Mat}(k, n)$ is stratified by rank and the largest stratum in the complement of $\mathrm{St}(k, n)$ is given by rank $k-1$ matrices which has codimension $n-k+1$.

Corollary 1.7. If $\dim X < n-k$ or $\dim X < 2(n-k)+1$ then $[X, \mathrm{St}(k, n)] = *$. In particular $\mathrm{St}(k, \infty)$ and $\mathrm{St}^{on}(k, \infty)$ are contractible.

Thus we have stumbled upon the fact that $\mathrm{St}^{on}(k, \infty)$ is a contractible space with a free $O(k)$ or $U(k)$ action and $\mathrm{Gr}(k, \infty)$ is the quotient.

Definition 1.8. Let G be a topological group which is nice in the sense that it has the homotopy type of a CW complex. A CW complex with a principal G bundle $EG \rightarrow BG$ with contractible total space is called a **classifying space** for G .

Theorem 1.9. $EG \rightarrow BG$ is unique up to homotopy equivalence, canonical up to homotopy if you choose the base point $*$ in BG and identify the fiber over $*$ with G . Moreover, isomorphism classes of principal G -bundles over a CW complex X , $*$ with trivialization over $*$ are in bijection with homotopy classes of pointed maps $[X, BG]$.

In fact (2) implies (1) because we have identified BG as a functor out of the homotopy category of pointed CW complexes, and the functor is unique up to natural isomorphism.

Remark 1.10. G is connected if and only if BG is simply connected. Extreme cases are G is discrete, then EG is the universal cover of BG .

Remark 1.11. G discrete, X is connected. Then $[X, BG]$ is the set of homomorphisms $\pi_1(X) \rightarrow G$ up to conjugation, whereas $[X, BG]_* = \text{Hom}(\pi_1(X), G)$. The first is isomorphism classes of principal G -bundles, the second is isomorphism classes of principal G -bundles with a trivialization at the base point.

Next we have two models of $BU(k) \rightarrow BU(k+1)$. One is the obvious inclusion, and the other model gives it as a fiber bundle with fiber S^{2k+1} .

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Last time we showed that $\text{Gr}_C(n, \infty)$ is a model for $BU(n)$. Note that there are parallel stories for $O(n)$ over $\mathbb{R}$ and $\text{Sp}(n)$ over $\mathbb{H}$ but we will now live over $\mathbb{C}$.

The Grassmannian $\text{Gr}(n, \infty)$ classifies iso classes of rank n complex vector bundles, and we have a principal $U(n)$ -bundle $\text{St}^{on}(n, \infty) \rightarrow \text{Gr}(n, \infty)$ which is the universal principal $U(n)$ -bundle with total space contractible.

We also know that we have embeddings $\text{Gr}(n, \infty) \rightarrow \text{Gr}(n+1, \infty)$ which is highly connected. One can think of inductively building $BU(n+1)$ from $BU(n)$ by attaching an extra cell of real $\dim 2n+2$ ($n+1, 4n+4$ in the real and quaternionic cases) at each step.

Example 2.1. If $n = 0$ then $\text{Gr}(1, \infty) = k\mathbb{P}^\infty$. $\mathbb{H}\mathbb{P}^\infty$ is $BSU(2)$ and the homotopy equivalence $SU(2) \hookrightarrow \text{GL}_1(\mathbb{H})$ shows that $SU(2)$ bundles are the same as $\mathbb{H}$ -line bundles.

Observe that $U(n) \hookrightarrow U(n+1)$ has quotient S^{2n+1} from the action of $U(n+1)$ on the unit sphere in $\mathbb{C}^{n+1}$. This gives a fiber bundle $S^{2n+1} \rightarrow BU(n) \rightarrow BU(n+1)$ which is the other model of the inclusion $BU(n) \rightarrow BU(n+1)$.

Proposition 2.2. The total space of the fiber bundle $\text{St}^{on}(n+1, \infty) \times S^{2n+1} \rightarrow \text{Gr}(n+1, \infty)$ is homotopy equivalent to $\text{Gr}(n, \infty)$ and the fiber bundle map is homotopic to the inclusion $\text{Gr}(n, \infty) \rightarrow \text{Gr}(n+1, \infty)$. Note that there is a natural lift $\text{Gr}(n, \infty) \rightarrow \text{St}^{on}(n+1, \infty)$ which is a homotopy equivalence.

This gives a nonprincipal fiber bundle $S^{2n+1} \rightarrow \text{Gr}(n, \infty) \rightarrow \text{Gr}(n+1, \infty)$.

Proof. A map $X \rightarrow \text{St}_C(n+1, \infty) \times_{U(n+1)} S^{2n+1}$ classifies up to isomorphism a complex vector

bundle of rank $n + 1$ together with a section of unit norm. This is equivalent (up to isomorphism and rescaling of hermitian metrics (which is a contractible space)) to $E^{n+1} \rightarrow X, s$ where s is a nonvanishing section. This is equivalent to a rank n vector bundle by taking the quotient. Thus the target space $\text{St}_{\mathbb{C}}(n + 1, \infty) \times_{U(n+1)} S^{2n+1}$ is a classifying space for rank n vector bundles, and thus homotopy equivalent to $\text{Gr}(n, \infty)$. The map to the base $\text{Gr}(n + 1, \infty)$ classifies the stabilized bundle of rank $n + 1$ and so is homotopy equivalent to natural map. $\square$

Now let us consider the general construction, note that $S^{2n+1} = U(n + 1)/U(n)$ is a homogeneous space. Let G be a nice topological group. Let H be a closed subgroup of G .

Let $EG \rightarrow BG$ be the classifying space for principal G -bundles. Suppose that $H \rightarrow G \rightarrow G/H$ is a principal H -bundle, in particular locally trivial. This is always true for Lie groups.

Theorem 2.3. EG/H is a classifying space for H . In particular, in the diagram below, there is a unique up to homotopy lifting of this map which is a homotopy equivalence.

$$\begin{array}{ccc} EG & \xrightarrow{f} & EG/H \\ p \downarrow & \nearrow h & \downarrow q \\ BH & \longrightarrow & BG \end{array}$$

Proof. First we need a lemma. Isomorphism classes of principal H -bundles on X are the same as isomorphism classes of principal G bundles together with a section of associated G/H -bundle up to homotopy of sections. This is saying that if $P \rightarrow X$ is a principal H -bundle, then you can form $Q = P \times_H G$ which is a principal G -bundle. A section of Q/H is the same as an isomorphism of Q with a bundle of the form $P \times_H G$. $\square$

Now we can build classifying spaces for all Lie groups. Recall from the Peter Weyl theorem that every compact Lie group has a faithful finite dimensional unitary representation, faithful being the key word here. This gives us $G \hookrightarrow U(n)$ for some n . So we can build a classifying space for G using our favorite model for $BU(n)$ and then dividing by G

$$EU(n) \rightarrow EU(n)/G = BG$$

and the good thing here is that this makes BG into an increasing union of compact manifolds which stabilize in homology. This is the story of classifying spaces for compact Lie groups.

Remark 2.4 (Slice theorem for quotients). When groups act on spaces, you have to be careful when you take quotients. It might not be Hausdorff for example. Let G be a possibly noncompact Lie group acting properly on a normal Hausdorff space X . (Normal means that have separation of closed sets by open neighborhoods. Urysohn's lemma says that this means we can separate by values of continuous functions.) Then there is a slice through each point: for each $x \in X$, there is a G_x -invariant

neighborhood S of x such that the natural map $G \times_{G_x} S \rightarrow X$ is an equivariant homeomorphism onto a G -invariant neighborhood of the orbit $G \cdot x$. This gives a linear local model using the normal space at the orbit. **insert more remark**

2.1 Three classical theorems

Theorem 2.5 (Leray Hirsch). Let $F \rightarrow E \xrightarrow{p} B$ be a connected fiber bundle over a CW base. Let R be a coefficient ring for cohomology, i.e a quotient of $\mathbb{Z}$ or something between $\mathbb{Z}$ or $\mathbb{Q}$. Suppose $H^*(F; R)$ is a free R -module on $\alpha_1, \dots, \alpha_k$ and that there exist classes $a_1, \dots, a_k \in H^*(E; R)$ such that $a_i|_F = \alpha_i$. **Note that this is a strong assumption because there is some subtlety about identification of F with the fiber over a point in B .**

Then the map $H^*(F; R) \otimes_R H^*(B; R) \rightarrow H^*(E; R)$ given by $x \otimes y \mapsto p^*y \cup a_i$ is an isomorphism of R -modules.

Remark 2.6. This recovers Kunneth formula for the product $E = F \times B$ by taking $a_i = \alpha_i \otimes 1$. If F has R -torsion, this cannot work because of Kunneth.

Example 2.7 (Nonexamples). $\mathbb{R} \rightarrow S^1$. $S^3 \subset \mathbb{C}^2 \rightarrow S^2 = \mathbb{CP}^1$ the Hopf fibration. In the second case, we compute

$$\begin{aligned} H^*(S^3; \mathbb{Q}) &= \mathbb{Q} \oplus \mathbb{Q}[-3] \\ H^*(S^2; \mathbb{Q}) \otimes H^*(S^1; \mathbb{Q}) &= \mathbb{Q} \oplus \mathbb{Q}[-1] \oplus \mathbb{Q}[-2] \oplus \mathbb{Q}[-3] \end{aligned}$$

What is happening is that there is a grid

$$\begin{bmatrix} \mathbb{Z} & 0 & \mathbb{Z} \\ \mathbb{Z} & 0 & \mathbb{Z} \end{bmatrix}$$

and there is a differential d_2 from the $\mathbb{Z}$ in the bottom right to the $\mathbb{Z}$ in the top left which kills the $\mathbb{Z}$ in the bottom right and the $\mathbb{Z}$ in the top left.

Proof. Idea of proof. Build B from its skeleta B_n and check that the map remains an iso at each step from the LES and the five lemma.

$$H^{*-1}(B_n) \rightarrow H^*(B_{n+1}, B_n) \rightarrow H^*(B_{n+1}) \rightarrow H^*(B_n) \rightarrow H^{*+1}(B_{n+1}, B_n)$$

Tensor with $H^*(F)$. Also have a sequence for $\pi^{-1}(B_n)$. Get a map from $H^*(F) \otimes H^*(B_n)$ to $H^*(\pi^{-1}(B_n))$ and check that it is an iso at each step. The key input is that we have a map between the original complexes. $\square$

Variant with the same proof: $E' \hookrightarrow E$ two fiber bundles over B . Use $H^*(F, F'; R)$ and $H^*(E, E'; R)$. Same hypothesis and conclusions.

Example 2.8. $U(n-1) \rightarrow U(n) \rightarrow S^{2n-1}$ will work, even though $U(n)$ is not homotopy equivalent to a product of odd spheres. Rationally it is, but integrality is the issue.

The second classical theorem is an application of Leray.

Theorem 2.9 (Thom isomorphism theorem). Let $E^r \rightarrow X$ be a real vector bundle of rank r with a metric. Let $D(E)$ be the unit disk bundle and $\partial D(E)$ be the unit sphere bundle. **Assume** that E is relatively orientable for a ring R . Then there exist a class $\theta \in H^r(D(E), \partial D(E); R)$ which restricts to the preferred generator in each fiber. Consequently by Leray, θ is such that the map $H^*(X; R) \rightarrow H^{*+r}(D(E), \partial D(E); R)$ given by $x \mapsto x \cup \theta$ is an isomorphism.

E is relatively orientable means that we can choose orientations of the fibers which vary continuously over the base. If $R = \mathbb{Z}/2$ there is no condition. If X is an oriented manifold then $(D^r, \partial D^r)$ becomes an oriented manifold with boundary, and then E is Poincare dual to the zero section of E . θ is called the Thom class of E and the isomorphism is called the Thom isomorphism.

The theorem is ok if you just have disk bundles and its boundary. One has disk bundles which do not come from vector bundles. The third theorem is the Gysin sequence but we will not get there today. This will show that $H^*(\text{Gr}(n, \infty))$ is a polynomial ring on n generators of degree $2, 4, \dots, 2n$ called the Chern classes.

Definition 2.10. The Euler class of E^r is the image of θ under the map $H^r(D(E), \partial D(E); R) \rightarrow H^r(D(E); R) = H^r(X; R)$ given by inclusion.

If X is compact oriented manifold, then $PD(e)$ is the zero locus of a generic section (transversal intersection with the zero section of E).

2.2 Line bundles and chern classes

Recall isomorphism classes of real, complex, quaternionic line bundles are classified by $BO(1) = \mathbb{RP}^\infty$, $BU(1) = \mathbb{CP}^\infty$, $B\text{SU}(2) = \mathbb{HP}^\infty$ respectively, and the cohomologies are (use the positive generators)

$$\begin{aligned} H^*(\mathbb{RP}^\infty; \mathbb{Z}/2) &= \lim_{n \rightarrow \infty} H^*(\mathbb{RP}^n; \mathbb{Z}/2) = \mathbb{Z}/2[w], \deg w = 1 \\ H^*(\mathbb{CP}^\infty; \mathbb{Z}) &= \lim_{n \rightarrow \infty} H^*(\mathbb{CP}^n; \mathbb{Z}) = \mathbb{Z}[c], \deg c = 2 \\ H^*(\mathbb{HP}^\infty; \mathbb{Z}) &= \lim_{n \rightarrow \infty} H^*(\mathbb{HP}^n; \mathbb{Z}) = \mathbb{Z}[p], \deg p = 4 \end{aligned}$$

Proposition 2.11. w, c are the Euler classes of the respective line bundles over $\mathbb{RP}^\infty$ and $\mathbb{CP}^\infty$ respectively. For $\mathbb{CP}^\infty$, we use $\mathcal{O}(1)$ the dual of the tautological sub line bundle.

Proof. A linear section has zero locus a hyperplane, the PD of the positive generator of cohomology. $\square$

Definition 2.12 (Characteristic classes). For a real line bundle $L \rightarrow X$ classified by $k_L : X \rightarrow \mathbb{RP}^\infty$, the Stiefel-Whitney class of L is $w(L) = k_L^* w$. For a complex line bundle $L \rightarrow X$ classified by $k_L : X \rightarrow \mathbb{CP}^\infty$, the Chern class of L is $c(L) = k_L^* c$.

Theorem 2.13. $w_1(L) = w(L)$ and $c_1(L) = c(L)$ determine the isomorphism class of L . Moreover,

$$\begin{aligned} w_1(L \otimes L') &= w_1(L) + w_1(L') \\ c_1(L \otimes L') &= c_1(L) + c_1(L') \end{aligned}$$

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3.1 Suspensions and smash product

You should think of smash product as a tensor product. This becomes precise in linearized, i.e. stable homotopy theory. Also we need base points to talk about smash products.

Example 3.1. $I, 0 \wedge I, 0$ is not very interesting. $S^m \wedge S^n$ is S^{m+n} . To see this, identify $S^m, *$ with the unit cube $I^m, \partial I^m$ so that

$$\begin{aligned} S^m \wedge S^n &= (I^m \wedge I^n) / (\partial I^m \wedge I^n \cup I^m \wedge \partial I^n) \\ &\cong I^{m+n} / \partial I^{m+n} \cong S^{m+n} \end{aligned}$$

Definition 3.2. The reduced suspension of a pointed space $X, *$ is $\Sigma X = S^1 \wedge X$. If X does not have a base point, we can add one by taking the disjoint union with a point and declaring that to be the base point, the space X_+ . Then declare the unreduced suspension ΣX to be the reduced suspension of X_+ .

To think of a suspension, you can think of a double cone on X and collapse a copy of X that stretches from the top of the cone to the bottom.

Theorem 3.3. X based space. Then G -bundles on ΣX with a trivialization at the based point are in bijection with based homotopy classes of maps $[X, G]_*$.

Proof. Bundles are trivial on the cones over X . Only info is the gluing along $X, *$ which is given by a map $X \rightarrow G$ up to homotopy. $\square$

Corollary 3.4. $[\text{Bun}_G(\Sigma X)]_* = [\Sigma X, BG]_* = [X, \Sigma BG]_*$

Recall that $\text{Map}_*(\Sigma X, Y) \cong \text{Map}_*(X, \Omega Y)$ where ΩY is the based loop space of Y if X is Hausdorff.

Corollary 3.5. The classifying map of the standard G bundle $\Sigma G \rightarrow BG$ induces a homotopy equivalence $G \rightarrow \Omega BG$ which preserves the group structure up to homotopy.

3.2 Line bundles on spheres

For $n > 0$ then we can compute

$$[\text{Bun}_{\text{GL}(1)}(S^n)]_* = [S^{n-1}, \text{GL}(1)]_*$$

Note that $[S^{n-1}, \pm 1] = \pm 1$ if $n = 1$ and $*$ if $n > 1$. Also note that $[S^{n-1}, S^1] = \mathbb{Z}$ if $n = 2$ and $*$ if $n \neq 2$. This is because any S^{n-1} with $n > 2$ is simply connected so maps lift to $\mathbb{R}$, the universal cover of S^1 , which is contractible.

Remark 3.6. The Eilenberg-MacLane space of type π, n where π is an abelian group and n is an integer is the unique up to homotopy, connected based space $K(\pi, n)$ such that $\pi_n(K(\pi, n)) = \pi$ and $\pi_m(K(\pi, n)) = 0$ for $m \neq n$.

We have met a couple examples of Eilenberg-MacLane spaces: $K(\mathbb{Z}/2, 1) = \mathbb{RP}^\infty$ and $K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$. S^1 is a $K(\mathbb{Z}, 1)$. Note that $\mathbb{HP}^\infty$ is not a $K(\mathbb{Z}, 4)$ because there are higher homotopy groups. A cool example is that $K(\mathbb{Z}/n, 1)$ is the total space of the sphere bundle of $\mathcal{O}(1)^n$ over $\mathbb{CP}^\infty$, also known as $S^\infty/\mathbb{Z}/n$ where $\mathbb{Z}/n$ acts by n -th roots of unity on the unit sphere in $\mathbb{C}^\infty$.

Theorem 3.7. If X is a CW complex, then $[X, K(\pi, n)]_* = H^n(X; \pi)$. This implies that $K(\pi, n)$ is unique up to homotopy, i.e. $K(\pi, n)$ is canonical in the homotopy category. When $n = 1$ we get that

$$[X, K(\pi, 1)]_* = \text{Hom}(\pi_1(X), \pi)$$

Moreover the equality is functorial in X and is induced by $f : X \rightarrow K(\pi, n)$ goes to $f_*\iota_n$ where ι_n is the distinguished class in $H^n(K(\pi, n); \pi)$, classified by the identity map $K(\pi, n) \rightarrow K(\pi, n)$. Removing the basepoints divides by the conjugation action of π on $\text{Hom}(\pi_1(X), \pi)$.

$$[X, K(\pi, n)] = \prod_{\text{connected components}} \text{Hom}(\pi_1(X_c), \pi) / \text{Ad}(\pi)$$

Corollary 3.8. $\text{Pic}_{\mathbb{R}}(X) \cong H^1(X; \mathbb{Z}/2)$ and $\text{Pic}_{\mathbb{C}}(X) \cong H^2(X; \mathbb{Z})$ given by $L \mapsto w_1(L)$ and $L \mapsto c_1(L)$ respectively.

Beware that there is a choice of sign in the complex case. Note that for us, the distinguished class $\iota_2 \in H^2(\mathbb{CP}^\infty; \mathbb{Z})$ is the Chern class of the dual $\mathcal{O}(1)$ of the tautological line bundle $\mathcal{O}(-1)$.

Remark 3.9. ΣG to BG reduces to $S^2 = \Sigma S^1$ to $\mathbb{CP}^\infty$ but we have to be careful about the orientation, i.e. the choice of $\theta \mapsto e^{i\theta}$ in the S^1 which is getting suspended actually corresponds to $\mathcal{O}(-1)$. This is because we get z which has a pole at infinity, whereas $\mathcal{O}(1)$ has a zero infinity.

Proof. Proof that homotopy classes of maps $[X, \mathbb{CP}^\infty]_* = H^2(X; \mathbb{Z})$. The proof is by induction on skeleta. Let X_n be the n -skeleton of X . Write $X_2 = \vee S^2$ because $\mathbb{CP}^\infty$ is simply connected and so X_1 can be contracted. Cellular approximation says that we can just look at maps to $\mathbb{CP}^1$. We get one integer for each 2-cell. This defines a class in $H^2(X_2; \mathbb{Z})$. If that class is zero, we can contract that map from X_2 . How to go further?

Inductive step from X_n to X_{n+1} . Suppose first $n \geq 3$. Here $H^2(X_{n+1}, \mathbb{Z}) \rightarrow H^2(X_n; \mathbb{Z})$ is an isomorphism. Need to show that maps to $\mathbb{CP}^\infty$ extend uniquely. But this is true because the map $S^n \rightarrow \mathbb{CP}^\infty$ is nullhomotopic for $n \geq 3$ and so the map will extend across all disks that we are attaching to get X_{n+1} .

The question is could there be 2-extensions? No. Can deform the guaranteed homotopy to keep it fixed on S^n . This is a topological argument. You have to squint just right.

Now consider $n = 3$. Have a map $H^2(X_3; \mathbb{Z}) \rightarrow H^2(X_2; \mathbb{Z})$ which we assume is surjective, i.e. for every $\omega_2 \in H^2(X_2; \mathbb{Z})$ there is a $\omega_3 \in H^2(X_3; \mathbb{Z})$ which restricts to ω_2 . We know that for ϕ the attaching map of a 3 disk, we must have $\phi^* \omega_2 = 0 \in H^2(S^2; \mathbb{Z})$. Else, a differential out of ω_2 would block the extension. Induction shows that a map out of ∂D^3 was contractible in $\mathbb{CP}^\infty$ and so argue as before. $\square$

3.3 Additivity

We will give two arguments for the facts

$$\begin{aligned} w_1(L \otimes L') &= w_1(L) + w_1(L') \\ c_1(L \otimes L') &= c_1(L) + c_1(L') \end{aligned}$$

Complex case. Have two maps $\phi, \phi' : X \rightarrow \mathbb{CP}^\infty$ classifying L and L' and we want to understand the tensor product bundle. What is the tensor product? It is the pullback of $\mathcal{O}(1, 1) = \mathcal{O}(1) \boxtimes \mathcal{O}(1)$ on $\mathbb{CP}^\infty \times \mathbb{CP}^\infty$ along the map $\phi \times \phi' : X \rightarrow \mathbb{CP}^\infty \times \mathbb{CP}^\infty$. The map $t : \mathbb{CP}^\infty \times \mathbb{CP}^\infty \rightarrow \mathbb{CP}^\infty$ classifying $\mathcal{O}(1, 1)$. We need to show that $t^* c_1 = c$ where c is the first Chern class of $\mathcal{O}(1, 1)$. But then we compute

$$c_1(\mathcal{O}(1, 1)) \in H^2(\mathbb{CP}^\infty \times \mathbb{CP}^\infty; \mathbb{Z}) \cong H^2(\mathbb{CP}^\infty; \mathbb{Z}) \oplus H^2(\mathbb{CP}^\infty; \mathbb{Z})$$

agrees with $c_1 \oplus c_1$ because restrictions are $\mathcal{O}(1)$ on each factor. Thus

$$t \circ (\phi \times \phi')^* c_1 = (\phi \times \phi')^* (c_1 \oplus c_1)$$

$\square$

Remark 3.10. Behind the theorem we see that $\mathbb{CP}^\infty$ is up to homotopy a topological abelian group, and the tensor product of line bundles is given by the group structure, i.e. $\mathbb{CP}^\infty \cong \text{Sym}^\infty \mathbb{CP}^1$. For generalized cohomology theories $c_1(L \otimes L')$ defines a 1 parameter formal group, and this quantity $c_1(L \otimes L')$ very nearly determines the cohomology theory. This is due to Quillen and is now christened chromatic homotopy theory.

Definition 3.11. Let V over X be a complex vector bundle of rank r . Form $\mathbb{P}V$ the projectivization of V which is a fiber bundle π over X with fiber $\mathbb{CP}^{r-1}$. Over $\mathbb{P}V$ there is a hyperplane bundle $\mathcal{O}(1)$ which is the dual of the tautological line bundle. Let $\omega = c_1(\mathcal{O}(1))$. Observe that $1, \dots, \omega^{r-1}$ form a basis for $H^*(\mathbb{P}V_*, \mathbb{Z})$ over each fiber. By the Leray Hirsch theorem

$$H^*(\mathbb{P}V; \mathbb{Z}) \cong \bigoplus_{i=0}^{r-1} \pi^* H^i(X; \mathbb{Z}) \cup \omega^i$$

This implies that there exist a unique relation

$$\omega^r + c_1(V) \cup \omega^{r-1} + \dots + c_r(V) = 0$$

with $c_i(V) \in H^{2i}(X; \mathbb{Z})$. These are called the Chern classes of V . The sum

$$c(V) = 1 + c_1(V) + \dots + c_r(V)$$

is called the total Chern class of V .

- Theorem 3.12.**
1. For $V = L$ the new definition of c_1 agrees with the old one.
 2. For any V , the top Chern class $c_r(V)$ is the Euler class of V as a real bundle, i.e. the Thom class of E restricted to the zero section.
 3. The Chern classes detect complex conjugation, i.e. $c_i(\overline{V}) = (-1)^i c_i(V)$.

Proof. To check (1) note that $\mathbb{P}V = X$ and the equation is $\omega + c_1(L) = 0$ but $L = \mathcal{O}(-1)$ and so that is ok. Postpone (2) and (3). $\square$

3.4 Characteristic properties

- Theorem 3.13.**
1. The classes $c_k(V)$ are functorial in V , i.e. for $f : X \rightarrow Y$ with V a vector bundle over Y , we have $c_k(f^*V) = f^*c_k(V)$.
 2. $c(L) = 1 + c_1(L)$ for line bundles.

Whitney sum For $V = L_1 \oplus L_2$ a direct sum of line bundles, we have $c(V) = c(L_1) \cup c(L_2)$.

3. The above properties completely characterize the Chern classes, i.e. if $c'_k(V)$ is another collection of classes satisfying the above properties, then $c'_k(V) = c_k(V)$ for all k and all V .

Proof. Quick proof of (3). Think about the total space $\mathbb{P}(V \oplus W)$. Inside you have $\mathbb{P}V$ and $\mathbb{P}W$ as disjoint submanifolds. Moreover, observe that $\mathbb{P}(V \oplus W) \setminus \mathbb{P}V$ is a bundle over $\mathbb{P}W$ by projection in homogeneous coordinates. Thus we can compute

$$H^*(\mathbb{P}(V \oplus W)) \cong H^*_{\mathbb{P}W}(\mathbb{P}(V \oplus W))$$

cohomology with supports. Consider now the expression

$$\left(\sum c_i(V) \cup \omega^{r-i}\right) \cup \left(\sum c_j(W) \cup \omega^{s-j}\right)$$

where r, s are the ranks of V, W . The left is 0 in $\mathbb{P}(V)$ and the right is 0 in $\mathbb{P}(W)$, i.e. the left comes from $H^*_{\mathbb{P}W}(\mathbb{P}(V \oplus W))$ and the right comes from $H^*_{\mathbb{P}V}(\mathbb{P}(V \oplus W))$. Thus the cup product is supported nowhere and thus must be zero. But $\sum_m \omega^{r+s-m} \cup c_m(V \oplus W)$ was the only monic equation that could be zero, so we get the desired formula. $\square$

4 Feb 26

4.1 Cohomology of flag varieties

Any universal identity involving chern classes need only be checked on unordered direct sums of line bundles. Given $\pi : V^r \rightarrow X$ we get the full flag variety $\text{Fl}(V)$ which is a bundle over X whose points over x are full flags in the fiber V_x . Choose a hermitian metric. Over $\text{Fl}(V)$ we have a filtration of π^*V by complete flags with subquotients L_i which gets that $\pi^*(V) \cong \bigoplus L_i$. The metric is what lets us split the filtration, and so in algebraic geometry we cannot necessarily do this. Observe that $\text{Fl}(V)$ is a iterated fiber bundle of projective spaces of decreasing dimension.

Let $x_i \in H^2(\text{Fl}(V); \mathbb{Z})$ be the first Chern class of L_i . An application of Leray Hirsch implies the following

Theorem 4.1. $H^*(\text{Fl}(V); \mathbb{Z}) \cong H^*(X; \mathbb{Z}) \langle x_1^{i_1} \dots x_r^{i_r} \rangle$ where $0 \leq i_k \leq r - k$ as a vector space. As an algebra

$$H^*(\text{Fl}(V); \mathbb{Z}) \cong H^*(X; \mathbb{Z})[x_1, \dots, x_r] / \langle \prod (1 + x_i) = c(V) \rangle$$

There is an action of S^r on $H^*(\text{Fl}(V); \mathbb{Z})$ by permuting the x_i and $\pi^*H^*(X; \mathbb{Z}) \hookrightarrow H^*(\text{Fl}(V); \mathbb{Z})$ is the inclusion of the S^r -invariants.

Having a metric, $\text{Fl}(\mathbb{C}^r) \cong U(r)/U(1)^r$ where S^n acts on the right by permuting the columns.

Note that if $X = \text{pt}$, then $V \cong \mathbb{C}^r$ is just a fixed vector space and

$$\text{Fl}(V) = \text{Fl}_r.$$

The relation becomes

$$\prod_{i=1}^r (1 + x_i) = c(V) = 1$$

since all higher Chern classes of a bundle on a point vanish. Equating homogeneous parts gives

$$e_1(x) = e_2(x) = \cdots = e_r(x) = 0,$$

so the cohomology ring of the full flag variety is the coinvariant algebra of the symmetric group S_r :

$$H^*(\text{Fl}_r; \mathbb{Z}) \cong \mathbb{Z}[x_1, \dots, x_r] / (e_1, \dots, e_r) = \mathbb{Z}[x_1, \dots, x_r] / \langle \mathbb{Z}[x]_+^{S_r} \rangle,$$

For the relative flag bundle, the fiber part has cohomology given by the coinvariants, the base part by the invariants, and the total cohomology is the tensor product of the two.

4.2 Properties of the c_i

$c_k(V^\vee) = (-1)^k c_k(V)$ because we have already seen that the equality holds for c_1 and the splitting principle implies that it holds for all c_k .

The splitting principle also implies that the top Chern class $c_r(V)$ is the Euler class of V as a real bundle. It is true for line bundles and then the Euler class of $V \oplus W$ is the cup product of the Euler classes of V and W .

Proposition 4.2. The Euler class of $V \oplus W$ is the cup product of the Euler classes of V and W .

Proof. We show that it is true for the Thom classes, and recall that the Euler class is the restriction of the Thom class to the zero section. The Thom class is the unique class in $H^{\dim_{\mathbb{R}} V}(D(V), \partial D(V); \mathbb{Z})$ which restricts to the preferred generator in each fiber. V, W being complex have preferred orientation. Clearly $\Theta_v \cup \Theta_w$ satisfies the right property. $\square$

Aside, a priori we have $\Theta_v \cup \Theta_w$ is a class in $H^*(D(V) \times D(W), D(V) \times \partial D(W) \cup \partial D(V) \times D(W); \mathbb{Z})$ but this is isomorphic to $H^*(D(V \oplus W), \partial D(V \oplus W); \mathbb{Z})$. The Euler class can be explicitly represented using distributions and differential forms.

Remark 4.3 (Geometric argument for euler product formula). If X is a compact oriented manifold, then $PD(e(V))$ is the zero locus of a generic section of V , which can be made smooth by transversality. Then same for W and $V \oplus W$. The zero loci coincide after perturbation because cup product corresponds to intersection.

Remark 4.4. For a generic transverse section s ,

$$Z(s) \subset X$$

is a smooth codimension- k submanifold and

$$PD(e(V)) = [Z(s)] \in H_{n-k}(X).$$

Cup product corresponds to transverse intersection:

$$PD(e(V) \cup e(W)) = [Z(s_V)] \cap [Z(s_W)].$$

Now take $V = N_{X/Y}$. Suppose $i : X \hookrightarrow Y$ is an oriented embedding of codimension k , with normal bundle $N := N_{X/Y}$. There is a fundamental identity:

$$i^* i_*(\alpha) = \alpha \cup e(N) \quad \text{for } \alpha \in H^*(X).$$

This is the self-intersection formula.

The geometric meaning is as follows: i_* corresponds to pushing forward cycles from X into Y , and pulling back i^* amounts to intersecting again with X . The correction factor is the Euler class of the normal bundle. Thus

$$X \cdot X = e(N_{X/Y})$$

inside $H^*(X)$. So cupping with the Euler class of the normal bundle is precisely the operation of intersecting with X inside Y .

Stability of chern classes. Observe that $c_*(V \oplus \mathbb{C}) = c_*(V)$ because the extra factor is 1. In general vector bundles over X form a symmetric monoidal category under direct sum and tensor product. Iso classes form a commutative semiring, and we can take an additive group completion to a commutative ring $K^0(X)$, if X is compact and Hausdorff.

The objects in $K^0(X)$ are formal differences of stable isomorphism classes of finite rank vector bundles, where V and W are stably isomorphic if there exists some E such that $V \oplus E \cong W \oplus E$. This can be simplified since every E can be complemented to a trivial bundle, so V and W are stably isomorphic if there exists some n such that $V \oplus \mathbb{C}^n \cong W \oplus \mathbb{C}^n$. Then the Chern classes refine to a map

$$c_* : K^0(X) \rightarrow \prod H^{2k}(X; \mathbb{Z})$$

by defining $c_*(V - W) = c_*(V)/c_*(W)$. This is an exponential like map, turning sums into products.

Proposition 4.5. c_* is well defined and turns sums into products. Moreover, if X is compact, the a priori infinite series terminates.

Example 4.6. $X = \mathbb{CP}^\infty$ and consider the class $[-\mathcal{O}(1)] \in K^0(\mathbb{CP}^\infty)$. Then $c_*(-\mathcal{O}(1)) = 1/c_*(\mathcal{O}(1))$ which is $1/(1+c) = 1 - c + c^2 - c^3 + \dots$

c_* of a tensor product. Let x_i, y_j be the Chern roots of V, W respectively. Then the Chern roots of $V \otimes W$ are $x_i + y_j$. The proof is by the splitting principle.

4.3 Aside on algebraic geometry

In algebraic geometry, we study coherent sheaves. A sheaf S is coherent if locally, there is a presentation of S as the cokernel of a map of finite rank vector bundles.

It is a theorem that on a complex projective manifold, every coherent sheaf has a **finite** resolution by finite rank algebraic vector bundles. In fact, you can stop at the dimension of the manifold.

This is a combination of two theorems. Hilbert syzygy theorem is about doing this locally. The second theorem is about doing this globally, there are enough vector bundles by projectivity.

Use this to functorially define the Chern classes of a coherent sheaf S by taking a resolution by vector bundles and then taking the alternating product of the Chern classes of the vector bundles in the resolution. The result is independent of the choice of resolution. This feeds into the Hirzebruch Riemann Roch theorem.

4.4 Equivariant cohomology and equivariant Euler classes

X space, G compact Lie group acting on X . Can form the Borel construction $X_G = EG \times_G X$ which is a fiber bundle over BG with fiber X . Put $H_G^*(X) = H^*(X_G)$.

Extreme special cases. If G acts trivially on X , then $X_G = BG \times X$ and if BG has no torsion in cohomology, then $H_G^*(X) = H^*(BG) \otimes H^*(X)$.

If G acts freely on X , then $X_G = X/G$ and so $H_G^*(X) = H^*(X/G)$.

Application. If $\pi : V \rightarrow X$ is a vector bundle of rank r , it carries a natural $U(1)$ action on the fibers.

Proposition 4.7. The equivariant Euler class $e_{U(1)}(V) \in H_{U(1)}^*(X)$ is

$$\sum_{i=0}^r c_i(V) \cup \omega^{r-i}$$

where ω is the generator of $H_{U(1)}^*(\text{pt}) = H^*(\mathbb{CP}^\infty)$.

Proof. V becomes a vector bundle over $X \times \mathbb{CP}^\infty$ by tensoring the pullback with $\mathcal{O}(1)$. The equivariant Euler class is simply the Euler class of this new bundle, a class in $H^*(X \times \mathbb{CP}^\infty)$. Now use the formula for the Chern roots of a tensor product. $\square$

4.5 Hirzebruch Riemann Roch

Note that this generalizes to the Atiyah-Singer index theorem. Let X projective algebraic complex manifold and S a coherent sheaf on X . Denote by $\chi(X, S) = \sum (-1)^i \dim H^i(X, S)$ the Euler characteristic of S . Let t_i denote the Chern roots of the tangent bundle TX , and let s_i be the Chern roots of S . Define the Chern character of S to be

$$\text{ch}(S) = \sum e^{s_i} = \sum_{k=0}^{\infty} \frac{1}{k!} \sum s_i^k$$

in rational cohomology. Define the Todd class of X to be

$$\text{td}(X) = \prod \frac{t_i}{1 - e^{-t_i}} = \prod \frac{t_i}{1 - (1 - t_i + t_i^2/2! - t_i^3/3! + \dots)} = \prod \frac{t_i}{t_i - t_i^2/2! + t_i^3/3! - \dots}$$

Then

$$\chi(X, S) = \int_X \text{ch}(S) \cup \text{td}(X)$$

5 March 3

Today we will do integration along fibers of a bundle. Application will be segre calsses, transgressions of classes, cohomology of $BU(n)$ and $U(n)$, cohomology of $\text{Gr}(k, n)$, and Bott periodicity.

5.1 Integration along the fiber

This is also known as a transfer map, anytime the map goes the wrong way. Let M be compact oriented manifold, then we have a map called capping with the fundamental class $[M]$

$$\int_M : H^*(M) \rightarrow H^{*-n}(\text{pt})$$

where n is the dimension of M and this goes the wrong way for cohomology because it goes the same way as the map $M \rightarrow *$.

We can do this for any fiber bundle $\pi : E \rightarrow B$ with compact oriented fibers of dimension d and the fibers are oriented in a locally constant way.

Theorem 5.1. There is a map $\pi_* : H^*(E) \rightarrow H^{*-d}(B)$ where d is the dimension of the fibers, which is functorial in pullback, i.e. given a pullback square

$$\begin{array}{ccc}
E' & \longrightarrow & E \\
\downarrow & & \downarrow \pi \\
B' & \longrightarrow & B
\end{array}$$

we have $\pi'_* f^* = f^* \pi_*$, and satisfies the projection formula

$$\pi_*(\pi^* \alpha \cup \beta) = \alpha \cup \pi_* \beta$$

Additional functoriality comes in layers of fiber bundles. If $F' \rightarrow E' \rightarrow B'$ is a fiber bundle and $F \rightarrow E \rightarrow B$ is a fiber bundle and we have a map of fiber bundle, then we have $(\pi \circ \pi')_* = \pi_* \circ \pi'_*$.

When B is a manifold, this plays well with Poincare duality. Before we define the map, first we give a partial generalization when B is also a manifold. Note that the transfer map could be defined using Poincare duality on both sides and the pushforward of homology classes. This needs not the fiber bundle structure. Let $f : E \rightarrow B$ be a proper map of oriented connected manifolds. Note the fibers need not be manifolds. Then there is a functorial $f_* : H^*(E) \rightarrow H^{*-d}(B)$ where $d = \dim E - \dim B$ which extends the previous construction when f is a fiber bundle.

Proof. The map is simply the composition

$$\begin{aligned}
H^*(E) &\xrightarrow{PD} H_{\dim E - *}(E) \\
&\xrightarrow{f_*} H_{\dim E - *}(B) \\
&\xrightarrow{PD^{-1}} H^{*-d}(B)
\end{aligned}$$

where the middle map is the pushforward of homology classes. This relies on the base to be a manifold. Remark that this fails if B is not oriented, then it can be salvaged by twisted coefficients in the orientation sheaf provided f is relatively oriented, meaning $f^*(or_B) \cong or_E$. We can give a second construction by following the map as an embedding followed by projection. If B is finite dimensional, then we can factor f as an proper embedding $E \hookrightarrow B \times \mathbb{R}^N$ followed by projection to B for some large N . Just use Whitney embedding theorem to get the embedding and then use the projection. Then just have to define the pushforward for proper embedding of manifolds. Recall the tubular neighborhood theorem. If $i : X \rightarrow Y$ is a proper embedding then there exists $X \hookrightarrow N_{X/Y}$ the normal bundle of X in Y and an open embedding $N_{X/Y} \hookrightarrow Y$ which is the identity on X . The oriented assumption implies that this is oriented. Then Thom implies that we have a Thom isomorphism $H^*(X) \cong H^{*+\dim N_{X/Y}}(D(N_{X/Y}), S(N_{X/Y}))$. Handle the projection using Kunneth formula. $\square$

Remark that the transfer map can operate on the level of differential forms if we have a submersion and the fibers don't pose singularities.

Remark 5.2. Anytime Leray Hirsch holds, then $H^*(E) \cong H^*(B) \otimes H^*(F)$ and the map π_* just integrates the fiber direction. For example, if $u = \sum_i \pi^*(\alpha_i) c_i$ with $c_i|_F$ a basis of $H^*(F)$, then

$$\pi_*(u) = \sum_i \alpha_i \int_F c_i|_F$$

In this case, the copy of $H^*(B)$ sits inside $H^*(E)$ as the part with fiber degree 0. Fiber integration projects onto that part by pairing with the top fiber class.

For the flag bundle, this becomes:

$$H^*(\text{Fl}(V)) \cong \bigoplus_I \pi^* H^*(X) \cdot m_I(x_1, \dots, x_r),$$

and π_* extracts the coefficient of the unique monomial m_I representing the top class of the flag fiber. For $\text{GL}(n)$, the flag fiber is Fl_n and the top class is $x_1^{n-1} x_2^{n-2} \cdots x_{n-1}$. This can also be expressed symbolically as the algebraic operation of taking a polynomial $f \in H^*(X)[x_1, \dots, x_n]$ and applying

$$\pi_*(f) = \text{coefficient of } x_1^{n-1} x_2^{n-2} \cdots x_{n-1} \text{ in } f = \Delta^{-1} \sum_{\sigma \in S_n} \text{sgn}(\sigma) f(x_{\sigma(1)}, \dots, x_{\sigma(n)})$$

where $\Delta = \prod_{i < j} (x_i - x_j)$ is the Vandermonde determinant. Note that the sum is alternating and every alternating polynomial is divisible by Δ , whose quotient is a symmetric polynomial (and hence a pullback from the base).

5.2 Application: segre class of a vector bundle

Let $\pi : V \rightarrow X$ be a vector bundle and $\mathbb{P}\pi : \mathbb{P}V \rightarrow X$ be the projectivization of V . We have the hyperplane bundle $\mathcal{O}(1)$ on $\mathbb{P}V$ and its first Chern class ω .

Definition 5.3. The Segre class of V is defined to be

$$s(V) = \mathbb{P}\pi_* \left(\frac{1}{1 - \omega} \right) = \mathbb{P}\pi_*(1 + \omega + \omega^2 + \dots)$$

Theorem 5.4. $s(V) = c(V)^{-1}$, i.e. the Segre class is the multiplicative inverse of the Chern class.

Remark 5.5. In algebraic geometry, can define s of a bundle of projective varieties (cone). In complex AG, things are real codimension 2, thus we get fundamental class and can use this to integrate down.

Proof. **The transfer map is always a module map over the cohomology of the base**, i.e. $f_*(f^* \alpha \cup \beta) = \alpha \cup f_* \beta$. This is also called the projection formula.

Given this, the theorem is equivalent to the identity

$$\mathbb{P}\pi_*\left(\frac{1}{1-\omega} \cdot \mathbb{P}\pi^*c(V)\right) = 1$$

But $1 - \omega$ is the total Chern class of the tautological line bundle $\mathcal{O}(-1)$. Tautological and complement $= \mathbb{P}\pi^*V$. Thus the term $\frac{1}{1-\omega} \cdot \mathbb{P}\pi^*c(V) = c_*(V/\mathcal{O}(-1))$ so it is equal to $1 + c_1 + \cdots + c_{r-1}$ where r is the rank of V . The pushforward of this is a constant. Now check it for $\mathbb{P}^{r-1}$ $\square$

5.3 Chern classes of $T\mathbb{P}^n$

What is $T\mathbb{P}^n$? At a point $L \in \mathbb{C}^{n+1}$, the tangent space is the infinitesimal deformations of L inside $\mathbb{C}^{n+1}$. If you draw the picture of a line close to another line, you see that it is the graph of a function from L to $\mathbb{C}^{n+1}/L$. Thus the tangent space is $\text{Hom}(L, \mathbb{C}^{n+1}/L)$. We compute $T\mathbb{P}^n = (\mathbb{C}^{n+1}/L) \otimes L^\vee$ and an exact sequence

$$0 \rightarrow \mathcal{O} \rightarrow \mathcal{O}(1)^{\oplus n+1} \rightarrow T\mathbb{P}^n \rightarrow 0$$

where the first map is the diagonal embedding. Now take Chern classes.

$$\begin{aligned} c(T\mathbb{P}^n) &= c(\mathcal{O}(1)^{\oplus n+1})/c(\mathcal{O}) \\ &= (1 + c)^{n+1} \end{aligned}$$

In particular $c_n(T\mathbb{P}^n) = \binom{n+1}{n}c^n = (n+1)c^n$ is the Euler class and we can check that

$$\int_{\mathbb{P}^n} c_n(T\mathbb{P}^n) = n+1 = \chi(\mathbb{P}^n)$$

5.4 Cohomology of $\text{Gr}(k, n)$

We have bundles

$$0 \rightarrow T \rightarrow \underline{\mathbb{C}}^n \rightarrow Q \rightarrow 0$$

where T is the tautological bundle of rank k and Q is the quotient bundle of rank $n - k$. We have classes $c_i(T)$ and $c_j(Q)$ and the identity

$$c(T) \cup c(Q) = 1$$

The first observation is that these generators are redundant. In fact, $c_i(Q)$ can be expressed in terms of $c_j(T)$ and vice versa, e.g. $c_1(Q) = -c_1(T)$. One might worry that if one throws away half of the generators, then one might lose the relations. The second observation is that this is no issue, because relations are recovered from the condition that $c(T) \cup c(Q) = 1$. The third observation is that there are no other relations.

Proof. There is a fiber bundle $\text{Fl}(k) \times \text{Fl}(n-k) \hookrightarrow \text{Fl}(n) \rightarrow \text{Gr}(k, n)$ so

$$H^*(\text{Gr}(k, n)) = [\mathbb{Z}[x_i]/(e_i = 0)]^{S^k \times S^{n-k}}$$

The S_k invariants correspond to $c_i(T)$ and the S_{n-k} invariants correspond to $c_j(Q)$. $\square$

More generally for $\pi : V \rightarrow X$ and $\text{Gr } \pi : \text{Gr}(k, V) \rightarrow X$ the Grassmannian bundle, we have $H^*(\text{Gr}(k, V)) = H^*(X) \langle c_i(T), c_j(Q) \rangle / \langle c(T) \cup c(Q) = c(V) \rangle$.

5.5 Cohomology of $BU(n)$ and $U(n)$

Write

$$BU(n) = \text{Gr}(n, \infty)$$

from which it becomes clear that

$$H^*(BU(n); \mathbb{Z}) \cong \mathbb{Z}[c_1, \dots, c_n]$$

where c_i are the universal chern classes, in the sense that any vector bundle V over X is classified by a map $f : X \rightarrow BU(n)$ and $c_i(V) = f^*c_i$.

Now we can do transgression to get the cohomology of $U(n)$. Consider transgression on a mapping space with source M and target X . Produce cohomology classes in $H^*(\text{Map}(M, X))$ so called tautological classes. There is an evaluation map $\text{Map}(M, X) \times M \rightarrow X$. Given $\alpha \in H^*(X)$, we can pull it back to $\text{Map}(M, X) \times M$ and then use Kunneth to get components in $H^*(M)$.

Fun example where $X = BU(n)$ and $M = \Sigma$ is a closed surface. Get tautological classes in $H^*(\text{Map}(\Sigma, BU(n)))$ from c_i and a basis of $H^*(\Sigma)$.

Theorem 5.6 (Atiyah-Bott, Proceedings of Royal Society, 1980s, "Yang Mills"). In this case, the cohomology of $\text{Map}(\Sigma, BU(n))$ is freely generated by these classes as an algebra.

For us, consider the case $M = S^1, *$. Consider the maps

$$\begin{aligned} (S^1, *) \times \Omega BU(n) &\rightarrow BU(n) \\ S^1 \wedge U(n) &\rightarrow BU(n) \end{aligned}$$

where keep in mind $U(n) = \Omega BU(n)$ homotopy equivalent. We have the fiber bundle $U(n-1) \rightarrow U(n) \rightarrow S^{2n-1}$ where the last map is called q . Then

Theorem 5.7.

$$\int_{S^1} ev^* c_i = q^* vol_{S^{2n-1}}$$

Corollary 5.8. The classes in $H^{2k-1}(U(n))$ pulled back from $H^{2k-1}(S^{2k-1})$ are defined on any $U(n)$ for $n \geq k$. This implies the Leray assumption and we can compute $H^*(U(n))$ as the exterior algebra on the transgressions of the c_i .

6 March 10

6.1 Symmetric functions and grassmannians

There is a duality isomorphism

$$\begin{aligned} \mathrm{Gr}(k, n) &\cong \mathrm{Gr}(n - k, n) \\ (T \rightarrow \mathbb{C}^n \rightarrow Q) &\mapsto (\overline{T}^\perp \rightarrow \overline{\mathbb{C}^n} \rightarrow \overline{Q}^\perp) \end{aligned}$$

note that $T \mapsto T^\perp$ is complex antilinear, so we need to conjugate to get a complex linear map. The chern roots change sign.

Theorem 6.1. D is compatible with Schubert stratification. On labels, one exchanges 0s and 1s and reverses the sequence. On partitions, one takes the dual. If λ fits in the $k \times (n - k)$ box, then λ^D will fit in the $(n - k) \times k$ box.

Why do Schubert cells map to cells? We need to show that if $\dim(T \cap F_q)$ jumps at $q = i_1, i_2, \dots$, then $\dim(T^\perp \cap F_q^D)$ jumps at $n - i_1, n - i_2, \dots$. This is a linear algebra fact.

This duality is also witnessed by an algebraic duality in the ring of symmetric polynomials. Let R be the ring of symmetric polynomials in infinite variables.

Theorem 6.2. The map $s_\lambda \in R \mapsto PD(\Sigma_\lambda) \in H^{2|\lambda|}(\mathrm{Gr}(k, n))$ is a surjective ring homomorphism. The homogeneous variables are either the Chern roots of Q or the negative Chern roots of T . They are interchanged by the duality isomorphism and it interchanges the partition labels. In particular

$$PD(\Sigma_\lambda) \cdot PD(\Sigma_\mu) = \sum_{\nu} c_{\lambda\mu}^\nu PD(\Sigma_\nu)$$

are the Littlewood-Richardson coefficients, which coincide with the structure constants of the Schur functions. Using the negative Chern roots of T we can also write

$$\begin{aligned} c_i(T) &= (-1)^i e_i(x_1, \dots, x_k) = (-1)^i S_{(1^i)} \\ c_j(Q) &= s_j(x_1, \dots, x_k) = S_{(j)} \end{aligned}$$

where e_i is the i -th elementary symmetric function and s_j is the j -th complete symmetric function.

A proof of this theorem follows from the Pieri formula.

Theorem 6.3. The Pieri formula states that

$$c(Q) \cap [\Sigma_\lambda] = \sum_{\nu} [\Sigma_\nu]$$

where the sum is over ν obtained from λ by adding a horizontal strip, i.e. adding boxes to λ with no two in the same column.

Remark 6.4. Schur Weyl duality considers the two sided action of GL_k and S_n on $(\mathbb{C}^k)^{\otimes n}$.

Constantin doesn't want to do the proofs so we will just explain the logic. The geometric proof of Pieri. Look at $\text{Gr}(k, n)$ and understand the multiplication of $c_*(Q)$ on homogeneous polynomials. Pass to the flag bundle of Q over $\text{Gr}(k, n)$. Every Schubert variety Σ_λ has a rational section, regular over the Schubert cell. This is because Q can be trivialized over the Schubert cell, the basis coming from restricting the total coordinates of $C^n \rightarrow Q$ complementing the pivot coordinates. Use the identity $p_*(p^*c(Q) \cap \sigma_\lambda) = c(Q) \cap p_*\sigma_\lambda$ where upstairs you have a formula.

Remark 6.5. Similar idea can be used to prove that $[\Sigma_\lambda]$ is the Schur function s_λ in the Chern roots of Q . Use $\text{Fl}(k) \rightarrow \text{Fl}(T) \rightarrow \text{Gr}(k, n)$ with Chern roots ρ_i of T for $i = 1, \dots, k$. The fact is that $\text{Fl}(T)$ has a stratification lifting the one in $\text{Gr}(k)$. There are $k!$ strata over each Σ_λ , namely the strata of $\text{Fl}(k)$. $\text{Fl}(k)$ is an iterated fiber bundle of projective spaces. The fact is $PD(\Sigma_\lambda) = p_*PD(\rho_1^{\lambda_1+k-1} \cup \dots \cup \rho_k^{\lambda_k})$. The wrong way map in cohomology here becomes the correct map in homology, so we get the zero locus of maps generically bijective to Σ_λ .

Recall that we have the Jacobi-Trudi formula

$$\sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^k s_{\lambda_i+n-i} = \frac{A}{B}$$

where A is the determinant of the matrix with (i, j) -entry s_{λ_i+j-i} and B is A with $\lambda_i = 0$. Then the fact above is about the application of the Jacobi-Trudi formula to monomials.

Repeating the argument in K theory gives the character formula for representations of $\text{GL}(n)$ as Schur functions. We get a bunch of theorems

Theorem 6.6. 1. $H^0(\text{Fl}(k), L_\lambda) = S_\lambda$ is the highest weight representation of $\text{GL}(k)$ with highest weight λ .

2. (Bott improvement) $H^{>0} = 0$ if λ is dominant.

3. $\sum (-1)^i H^i$ coincides with the pushforward $p_!$ in K theory

4. $p_!$ in K theory obeys the same formula.

6.2 Bott periodicity theorem

We didn't give the definition of homotopy groups yet but we can give a weak version of the statement.

Theorem 6.7. $\pi_k(U) = 0$ for k even and $\pi_k(U) = \mathbb{Z}$ for k odd, where $U = \lim U(n)$ is the infinite unitary group.

Let's state the better version.

Theorem 6.8. There is a homotopy equivalence

$$\beta : BU \rightarrow \Omega^0 U$$

where $\Omega^0 U$ is the connected component of the based loop space of U containing the constant loop. In particular, $\Omega U \simeq \mathbb{Z} \times BU$.

Remark 6.9. Grouplike monoid structure of ΩU (any space honestly) means that $\Omega U \simeq \mathbb{Z} \times \Omega^0 U$ where $\mathbb{Z}$ is the fundamental group of U . To see this, note that as a space $U(n)$ is homeomorphic to $U(1) \times SU(n)$ the map to the first component being the determinant, and the map is split by $u \in \mathbb{R} \mapsto \text{diag}(u, 1, \dots, 1)$.

We can state the Bott map. Realize BU as limit of Grassmannian. Note that $\Omega SU(n)$ is maps $I \rightarrow SU(n)$ sending the endpoints to the identity. We can consider $P_{I,-I}(SU(2n))$ the space of paths in $SU(2n)$ sending the endpoints to $\pm I$. Note that $P_{I,-I}(SU(2n))$ is homotopy equivalent to $\Omega SU(2n)$ because we can simply choose any path from I to $-I$ and concatenate.

This receives a map from $\lim \text{Gr}(n, 2n)$. The map will go to minimal geodesics in $SU(2n)$. We will send $T \in \text{Gr}(n, 2n)$ as follows. Write $\mathbb{C}^{2n} = T \oplus T^\perp$ and define a path in $SU(2n)$ by sending $t \in I$ to $e^{\pi i t} \text{Proj}_T + e^{-\pi i t} \text{Proj}_{T^\perp}$.

Theorem 6.10. The image is a union of cells in a cell decomposition of a cellular approximation of $P_{I,-I}(SU(2n))$. The other cells has dimension $2n + 2$. In particular we get all the lower cells.

Sources of the cell decomposition.

1. Morse theory a la Bott.
2. Algebra. Coarsening of the Bruhat decomposition of ΩSU .

Part of the story is that there is an algebraic version of ΩU . We can consider the Laurent polynomial loop group and its flag variety. $L_{\text{poly}} \text{GL}_n(\mathbb{C})$ is the subgroup of invertibles in $\text{Mat}_n(\mathbb{C}[z, z^{-1}])$. Note that $U(n) \hookrightarrow \text{GL}_n(\mathbb{C})$ is a homotopy equivalence by the polar decomposition because M invertible implies that $M = up$ and P the positive definite part is contractible.

Theorem 6.11. $L_{poly} GL_n(\mathbb{C}) \hookrightarrow L GL_n(\mathbb{C})$ is a homotopy equivalence. The right hand side is a CW complex because you can filter by the order of zeros and poles.

We can get this from Bott or an analog of Schubert decomposition. We consider Bott's argument. Morse theory. Consider the energy functional on $P_{I,-I}(SU(2n))$ given by $E(\gamma) = \int_0^1 \|\gamma'(t)\|^2 dt$. The critical points are precisely the geodesics from I to $-I$. Note that $L_{sm} GL(n) \subset L_{cts} GL(n)$ is a homotopy equivalence by general nonsense so thinking about derivatives is harmless. The critical points are precisely the image of the map $\lim Gr(n, 2n) \rightarrow P_{I,-I}(SU(2n))$. Gradient flow defines a cell decomposition of $P_{I,-I}(SU(2n))$. The dimension of the cells is computable from the Hessian of E .

7 March 12

Characteristic classes for real bundles. Chern classes for G -bundles, i.e. $Sp(n), O(n), SO(n)$, etc. Why the argument for U does not work for O and SO . Main theorem about $H^*(BO, BSO; \mathbb{Z})$. Stiefel Whitney at $p = 2$. Pontrjagin away from $p = 2$. Proof for $BSO(n)$ and $\mathbb{Z}[1/2]$ coefficients. SW classes and Steenrod squares. Integral patching, the Bockstein homomorphism.

7.1

Recall G bundles on X are classified by homotopy classes of maps $X \rightarrow BG$, and the universal chern classes live in $H^*(BG)$. For example the cohomology of $BU(n)$ is a polynomial ring on the universal chern classes c_i .

A cheating method to define characteristic classes is to embed groups into $U(n)$ and pull back the chern classes. For example $Sp(n) \hookrightarrow U(2n)$ by splitting $\mathbb{H}$ as $\mathbb{C} \oplus \mathbb{C}j$. It turns out that you get nothing new as $H^*(BSp(n)) = \mathbb{Z}[c_2, c_4, \dots, c_{2n}]$ where c_{2i} is the pullback of the $2i$ -th chern class. In particular the chern roots come in pairs $x_i, -x_i$ and note that $1 + x_i$ and $1 - x_i$ multiply to $1 - x_i^2$.

There is analogous story for $O(n)$ and $SO(n)$ but we first must pass to the complexification $E \mapsto E \otimes \mathbb{C}$ to get a complex vector bundle and then take the chern classes.

We define $p_i(E) = (-1)^i c_{2i}(E \otimes \mathbb{C})$ to be the i -th Pontrjagin class. It lives in $H^{4i}(BO; \mathbb{Z})$ and vanishes after $i \geq n/2$. They satisfy an analogous Whitney sum formula, taken in $\mathbb{Z}[1/2]$ coefficients. We do not use c_{odd} because they are two torsion. Since $E \cong E^\vee$ as real bundles, their complexifications $E \otimes \mathbb{C}$ and $E^\vee \otimes \mathbb{C}$ are isomorphic as complex bundles, but they also have opposite complex structures so $c_i(E \otimes \mathbb{C}) = (-1)^i c_i(E^\vee \otimes \mathbb{C})$. Thus $c_{odd}(E \otimes \mathbb{C})$ are two torsion.

Remark 7.1. Sometimes can define an extra interesting characteristic class for SO bundles called the Euler class. Need SO because of orientability to define the Thom class. Can define this by looking at the disk bundle rel sphere bundle on the universal SO bundle over $BSO(n)$ and taking the Thom class. Thus we get a class $e_{2n} \in H^{2n}(BSO(2n))$ with the property that its pullback along any map

$X \rightarrow BSO(2n)$ is the Euler class of the corresponding bundle. This is because Thom classes are functorial. In $BSO(2n+1)$ we also have an Euler class but it is two torsion.

Remark 7.2. Defining the Euler class with $\mathbb{Z}$ coefficients requires orientability. It is always ok with $\mathbb{Z}/2$ coefficients. On a compact $\mathbb{Z}$ -oriented manifold M

$$\int_M e(TM) = \chi(M)$$

and an analogous formula holds for non-orientable manifolds with $\mathbb{Z}/2$ coefficients.

Remark 7.3. Another useful property of e . Say $p : E \rightarrow X$ is a fiber bundle with fibers compact oriented manifolds, $\dim d$. We have a relative Euler class for the vertical tangent bundle. Integration along the fiber gives a map $p_* : H^*(E) \rightarrow H^{*-d}(X)$. Precomposing with capping with the Euler class gives a map $H^{*-d}(X) \rightarrow H^*(E) \rightarrow H^{*-d}(X)$ which is **prop** multiplication by the Euler characteristic of the fiber. This is true because it's a module map, so follow where 1 goes.

Remark 7.4. Cohomology with coefficients. For nonorientable bundles E can define a class $\tilde{e}(E) \in H^n(X; or_E)$ where or_E is the orientation sheaf of E . Then can integrate the fiber M with twisted coefficients in the orientation sheaf to get $\int_M \tilde{e}(TM) = \chi(M)$. Morally this is about twisted functions along twisted measures, twisted in the sense that they both carry monodromy in the same way so that the monodromy cancels out and we get a well defined number.

Theorem 7.5. $H^*(BO; \mathbb{Z}/2) = \mathbb{Z}/2[w_1, w_2, \dots]$ where w_i is the i -th Stiefel Whitney class. They satisfy the Whitney sum formula

$$w(E \oplus F) = w(E) \cup w(F)$$

We also have the splitting principle, i.e. given $E \rightarrow X$, we can form the flag bundle $Fl(E) \rightarrow X$ and pull back E to $Fl(E)$, then it splits as a sum of line bundles. Moreover the pullback in cohomology is injective, so we can define the Stiefel Whitney classes by the formula

$$w(E) = \prod (1 + x_i)$$

where x_i are the Stiefel Whitney roots, i.e. the first Stiefel Whitney classes of the line bundles in the splitting. The pullback identifies $H^*(X)$ with the subalgebra of $H^*(Fl(E))$ invariant under the action of W . $w_1(E)$ is the obstruction to orientability of E .

$$H^*(BSO; \mathbb{Z}/2) = \mathbb{Z}/2[w_2, w_3, \dots]$$

In particular passing to orientability kills only w_1 . Analogously we have

$$H^*(BSU(n); \mathbb{Z}) = \mathbb{Z}[c_2, c_3, \dots, c_n]$$

kills only the first chern class, the obstruction to finding a section of the "orientation sheaf", which is the canonical bundle. Also fact: $w_n(E) = e(E) \in H^n(X; \mathbb{Z}/2)$.

Proposition 7.6. $c_1 \bmod 2$ is w_{2i} . For example $c_n \bmod 2$ is the Euler class mod 2, which is the top Stiefel Whitney class.

Postponed until later.

Example 7.7. $e(\mathbb{RP}^2) = 1 \bmod 2$. This is true but has nothing to do with Chern classes. What about $\mathbb{RP}^3$?

The main theorem says that these are all the classes, but we have to say how mod 2 and $\mathbb{Z}[1/2]$ fit together.

Theorem 7.8. Let n even. Then

$$H^*(BSO(n); \mathbb{Z}) \cong F \oplus T_2$$

where F is a free $\mathbb{Z}$ -module and T_2 is the 2-torsion. This will be split. T_2 is canonical, F is a quotient in general, but in this case, F will be a sub.

Also

$$H^*(BSO(n); \mathbb{Z}[1/2]) \cong \mathbb{Z}[1/2][\tilde{p}_1, \tilde{p}_2, \dots, \tilde{p}_{n/2-1}, e_n]$$

where we write $\tilde{p}_i$ to emphasize that they are pulled back from $BO(n)$ and e_n is the Euler class. We also have the top $p_{n/2}$ but we don't need to write it because it is the square of the Euler class.

Also we have

$$H^*(BO(n); \mathbb{Z}[1/2]) \cong \mathbb{Z}[1/2][p_1, p_2, \dots, p_{\lfloor n/2 \rfloor}]$$

subject to the relation $p_i = w_{2i}^2 \bmod 2$.

For n odd, $H^*(BSO(n); \mathbb{Z}[1/2]) = \mathbb{Z}[1/2][\tilde{p}_1, \tilde{p}_2, \dots, p_{\lfloor n/2 \rfloor}]$.

Remark 7.9. The p_i and $\tilde{p}_i$ are defined in the $\mathbb{Z}$ cohomology and the subring $\mathbb{Z}[\tilde{p}_1, \tilde{p}_2, \dots]$ (including e if even) is saturated in $H^*(BSO(n); \mathbb{Z})$. Equivalently the $p_i, \tilde{p}_i$ and e generate the torsion free part of the cohomology.

You might know that $\pi_1 SO(n) = \mathbb{Z}/2$ for $n \geq 3$ and so it has a double cover $\text{Spin}(n)$. If you look at the cohomology $H^*(B\text{Spin}(n); \mathbb{Z})$, you lose w_2 . But in $B\text{Spin}(n)$ p_1 becomes divisible by 4 if $n = 3$ and 2 if $n \geq 4$. This example is to show that these statements that hold for $BSO(n)$ are very delicate and good, that we do have all of the integral cohomology.

Let n even. We can give an inductive computation of $\mathbb{Z}[1/2]$ cohomology of $BSO(n)$ using the fiber bundle $SO(n+1)/SO(n) \rightarrow BSO(n) \rightarrow BSO(n+1)$ where the fiber is $S^n = S(E)$ where

$E = \mathbb{R}^{n+1}$ is the standard representation. Then study the Gysin sequence for the pair $D(E), S(E)$ and we get a long exact sequence with $\mathbb{Z}[1/2]$ coefficients.

$$\dots \rightarrow H^k(D(E), S(E)) \rightarrow H^k(D(E)) \rightarrow H^k(S(E)) \rightarrow H^{k+1}(D(E), S(E)) \rightarrow \dots$$

which can now be identified

$$\dots \rightarrow H^{k-n-1}(BSO(n+1)) \rightarrow H^k(BSO(n+1)) \rightarrow H^k(BSO(n)) \rightarrow H^{k-n}(BSO(n+1))$$

where the first map is capping with the Euler class and is in fact 0 since it is 2 torsion. Another fact is that the map $H^k(BSO(n)) \rightarrow H^{k-n}(BSO(n+1))$ is integration along the sphere fiber. So the image misses the Euler class, so we get what we are supposed to get.

n odd is similar but Constantin doesn't want to do it. BO is similar but there is no Euler class.

Remark 7.10. Stiefel Whitney classes are algebraically independent. But they are not independent cohomologically. In particular, there are universal cohomology operations with $\mathbb{Z}/2$ coefficients, functorial in X , called Steenrod power operations $Sq^i : H^*(X; \mathbb{Z}/2) \rightarrow H^{*+i}(X; \mathbb{Z}/2)$ and we have the total square

$$Sq = Sq^0 + Sq^1 + Sq^2 + \dots$$

which is an algebra homomorphism. The Wu relation gives a formula for the action of Sq on the Stiefel Whitney classes, in particular we can write it as quadratics in the Stiefel Whitney classes. "So only the w_{2k} are algebraically independent."

8 Missed classes

8.1 Homotopy groups

Definition 8.1. The n -th homotopy group of a pointed space (X, x) is the set of homotopy classes of maps $S^n \rightarrow X$ sending the base point to x . We denote it by $\pi_n(X, x)$ or simply $\pi_n(X)$ if X is path connected.

The group law is given by writing $S^n \rightarrow S^n \vee S^n$ as the pinch map and then composing with the wedge of two maps $S^n \rightarrow X$. $\pi_n(X, x)$ can also be modeled by the set of homotopy classes of maps $I^n \rightarrow X$ sending the boundary to x . In this model, the group law can be defined by concatenating along the 1st coordinate. The homotopy groups are abelian for $n \geq 2$.

There is an action of $\pi_1(X)$ on $\pi_n(X)$ for $n \geq 2$ by dragging the base point around loops. We say that X is **n -simple** if the action of $\pi_1(X)$ on $\pi_n(X)$ is trivial, and X is **simple** if it is n -simple for all n .

Proposition 8.2. A covering space projection $p : \tilde{X} \rightarrow X$ induces an isomorphism on π_n for $n \geq 2$.

Proof. The map is surjective because we can lift maps $S^n \rightarrow X$ to $\tilde{X}$ because S^n is simply connected. The map is injective because if a map $S^n \rightarrow \tilde{X}$ becomes nullhomotopic in X , then we can lift the nullhomotopy to $\tilde{X}$ and get a nullhomotopy there. $\square$

Moreover, if $\tilde{X}$ is the universal cover of X , then $\pi_1(X)$ acts on $\tilde{X}$ by deck transformations and this action induces the action of $\pi_1(X)$ on $\pi_n(\tilde{X}) \cong \pi_n(X)$.

Proposition 8.3. A product of path connected spaces X_α has homotopy groups given by the product of the homotopy groups of the factors, i.e. $\pi_n(\prod X_\alpha) \cong \prod \pi_n(X_\alpha)$.

Proof. This follows immediately from the universal property of products. $\square$

Another useful notion is the relative homotopy groups $\pi_n(X, A, x)$ for a pair of spaces $A \subset X$.

Definition 8.4. Let $x \in A \subset X$ be a pointed pair of spaces. The n -th relative homotopy group $\pi_n(X, A, x)$ is the set of homotopy classes of maps $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x)$ where

$$J^{n-1} = \overline{\partial I^n \setminus (I^{n-1} \times \{1\})}$$

with homotopies through such maps. Equivalently, it is homotopy classes of maps $(D^n, S^{n-1}, *) \rightarrow (X, A, x)$ modulo homotopies through such maps.

Theorem 8.5. There is a long exact sequence of homotopy groups

$$\cdots \rightarrow \pi_n(A) \rightarrow \pi_n(X) \rightarrow \pi_n(X, A) \rightarrow \pi_{n-1}(A) \rightarrow \cdots$$

coming from the inclusion of pairs. The boundary map is the restriction of a map $D^n \rightarrow X$ to the boundary $S^{n-1} \rightarrow A$.

8.2 Whitehead's theorem

Theorem 8.6. A map $f : X \rightarrow Y$ between CW complexes is a homotopy equivalence if and only if it induces isomorphisms on all homotopy groups.

Note that two CW complexes may have the same homotopy groups but not be homotopy equivalent. In particular, having a map inducing isomorphisms on homotopy groups is a much stronger condition than just having isomorphic homotopy groups. One situation where the latter implies the former is when all homotopy groups vanish, in which case the inclusion of a point is a homotopy equivalence.

Example 8.7. Consider $X = \mathbb{R}P^2$ and $Y = S^2 \times \mathbb{R}P^\infty$. These both have fundamental group $\mathbb{Z}/2\mathbb{Z}$. Their higher homotopy groups are isomorphic since their universal covers S^2 and $S^2 \times S^\infty$ are homotopy equivalent, S^∞ being contractible. But $\mathbb{R}P^2$ and $S^2 \times \mathbb{R}P^\infty$ are not homotopy equivalent since their homology groups are vastly different, $S^2 \times \mathbb{R}P^\infty$ having nonvanishing homology in infinitely many dimensions since it retracts onto $\mathbb{R}P^\infty$.

Example 8.8. Another pair of CW complexes that are not homotopy equivalent but have isomorphic homotopy groups is S^2 and $S^3 \times \mathbb{C}P^\infty$. Indeed, $\mathbb{C}P^\infty$ is an Eilenberg–MacLane space $K(\mathbb{Z}, 2)$, so

$$\pi_n(\mathbb{C}P^\infty) = \begin{cases} \mathbb{Z}, & n = 2, \\ 0, & n \neq 2. \end{cases}$$

Therefore

$$\pi_n(S^3 \times \mathbb{C}P^\infty) \cong \pi_n(S^3) \oplus \pi_n(\mathbb{C}P^\infty)$$

for all $n \geq 1$, and hence

$$\pi_n(S^3 \times \mathbb{C}P^\infty) \cong \begin{cases} \mathbb{Z}, & n = 2, \\ \pi_n(S^3), & n \geq 3. \end{cases}$$

Now consider the Hopf fibration

$$S^1 \longrightarrow S^3 \longrightarrow S^2.$$

Its long exact sequence in homotopy implies that

$$\pi_n(S^3) \cong \pi_n(S^2) \quad \text{for all } n \geq 3,$$

since $\pi_n(S^1) = 0$ for $n \geq 2$. Also,

$$\pi_2(S^2) \cong \mathbb{Z}, \quad \pi_1(S^2) = 0.$$

Since $S^3 \times \mathbb{C}P^\infty$ is simply connected as well, it follows that

$$\pi_n(S^3 \times \mathbb{C}P^\infty) \cong \pi_n(S^2) \quad \text{for all } n \geq 1.$$

However, these spaces are not homotopy equivalent. For example,

$$H_4(S^2) = 0,$$

while

$$H_4(S^3 \times \mathbb{C}P^\infty) \cong H_4(\mathbb{C}P^\infty) \cong \mathbb{Z},$$

so their homology groups are not isomorphic. Therefore S^2 and $S^3 \times \mathbb{C}P^\infty$ cannot be homotopy equivalent.

8.3 Cellular approximation and CW approximation

A map $f : X \rightarrow Y$ is **cellular** if it sends the n -skeleton of X to the n -skeleton of Y for all n .

Theorem 8.9. Any map $f : X \rightarrow Y$ between CW complexes is homotopic to a cellular map. If f is already cellular on a subcomplex $A \subset X$, then the homotopy can be taken to be relative to A .

It immediately follows that $\pi_k(S^n) = 0$ for $k < n$ because the k -skeleton of S^n is a point.

Definition 8.10. A map $f : X \rightarrow Y$ is a weak homotopy equivalence if it induces isomorphisms on all homotopy groups.

Whitehead's theorem reformulated says that a weak homotopy equivalence between CW complexes is a homotopy equivalence. However, a weak homotopy equivalence between general spaces need not be a homotopy equivalence. For example, there exist spaces which have trivial homotopy groups but are not contractible. For these spaces, the constant map to a point is a weak homotopy equivalence but not a homotopy equivalence.

Example 8.11. Consider the quasi-circle Y which is a closed subspace of $\mathbb{R}^2$ obtained by taking a portion of the graph of $y = \sin(1/x)$, the segment $[-1, 1]$ in the y -axis, and connecting them with a curve. This space comes with a map $p : Y \rightarrow S^1$ which collapses the sine curve and the interval to a point. Then this space has trivial homotopy groups. Any map from S^n to this space is nullhomotopic because the image is compact, so it must meet the sine-curve in finitely many oscillations. This will clearly be nullhomotopic. However, the quasi-circle is not contractible because if it were, then p would be nullhomotopic and we could lift p to the universal cover $\mathbb{R}$ of S^1 . The lifted map $\tilde{p} : Y \rightarrow \mathbb{R}$ in fact factors through $Y/I = S^1$ because the quotient map collapses I to a point. Then this gives a section of the universal cover $\mathbb{R} \rightarrow S^1$, which is impossible.

Definition 8.12. Let Y be a topological space. A *CW-approximation* of Y is a CW complex X together with a map $f : X \rightarrow Y$ such that for every CW complex Z , the induced map $f_* : [Z, X] \rightarrow [Z, Y]$ is a bijection, where $[-, -]$ denotes the set of homotopy classes of maps.

Taking $Z = S^n$ for all n shows that a CW approximation is a weak homotopy equivalence. Conversely, if $f : X \rightarrow Y$ is a weak homotopy equivalence and X is a CW complex, then f is a CW approximation by Whitehead's theorem.

Theorem 8.13. Every topological space Y admits a CW approximation $X \rightarrow Y$. If X is pathconnected, X can be chosen to have a single 0 cell, with all other cells attached by basepoint-preserving maps.

There is also a relative version of CW approximation.

Theorem 8.14 (Relative CW approximation). Let (X, A) be a pair of spaces with A a CW complex (or more generally a subspace). Then there exists a relative CW complex (Z, A) and a map of pairs

$$f : (Z, A) \longrightarrow (X, A)$$

such that:

1. $f|_A = \text{id}_A$,
2. f is a weak homotopy equivalence of pairs, i.e.

$$\pi_k(Z, A) \xrightarrow{\cong} \pi_k(X, A) \quad \text{for all } k.$$

If (X, A) is n -connected, i.e.

$$\pi_k(X, A) = 0 \quad \text{for all } k \leq n,$$

then there exists a relative CW complex (Z, A) and a weak equivalence of pairs

$$(Z, A) \xrightarrow{\cong} (X, A)$$

such that (Z, A) is obtained from A by attaching no cells in dimensions $\leq n$. Equivalently, the first relative cells occur in dimension $n + 1$.

9 Apr 7

Linking numbers and Hopf invariant. Proof of excision in homotopy. Freudenthal suspension theorem. Stable homotopy theory is linear in homotopy. The nonlinear structure in homotopy is the Whitehead bracket

$$[,] : \pi_k(X) \otimes \pi_l(X) \rightarrow \pi_{k+l-1}(X)$$

if $k, l > N$ disappears upon $2N$ -truncation. Nonlinear structure disappears as $N \rightarrow \infty$.

9.1 Linking numbers in S^N

Let M^p and N^q be closed oriented disjoint submanifolds of S^N with $p + q < N$. The interesting case is when $p + q = N - 1$ because if not, we can move them far apart so that they do not cross via ambient isotopy. This is the same as saying that we can enclose them in manifolds homeomorphic to disjoint balls. The argument is we can shift M to infinity in a general direction, and then a generic perturbation of a half-cylinder avoids N for dimension reasons. Dragging space along gets an isotopy.

This fails exactly when $p + q = N - 1$. In this case, we can define the linking number $\ell(M^p, N^q)$ as the intersection number of a cylinder on M with N . Independent because the difference of two

cylinders will be nullhomologous, i.e. a boundary. It can also be defined as the cup product of $PD(Cyl(M), M_\infty \cup M_0)$ and $PD(N)$ inside $H^{top}(S^N, \partial Cyl(M))$. Another way is to define a map $M \times N \rightarrow S^{N-1}$ by sending (m, n) to the intersection point of the ray through m and n with the sphere at infinity. Then the linking number is the degree of this map. Mind orientations, since in some dimensions, switching M and N changes the ray direction and thus the sign of the linking number.

Example 9.1. N odd so then $N-1$ is even. The antipodal map on S^{N-1} changes orientation. $M^p \times N^q \rightarrow N^q \times M^p$ has degree $(-1)^{pq}$ so if p, q odd there is no sign change. The example here is two circles in $\mathbb{R}^3$.

9.2 The Hopf invariant of $S^{4n-1} \rightarrow S^{2n}$

It is defined as the linking number of two generic fibers. It is also the self linking number of a single fiber, given a trivialization of the normal bundle, i.e. a framing. At a submersive point x , the map $df : N(f^{-1}(f(x))) \rightarrow T_{f(x)}S^{2n}$ is an isomorphism, so we can use it to trivialize the normal bundle.

In 3D TQFT, one can define invariants of three manifolds with embedded colored links. Physicists call them Wilson lines, and there are finite group reps living over these knots known as colors. Need trivialization of normal bundle to the link. The simplest example is abelian chern simons theory (abelian means for the torus). Linking numbers and self linking numbers are involved.

Theorem 9.2. The map $S^{4n-1} \rightarrow S^{2n}$ given by the Whitehead square of the identity map $S^{2n} \rightarrow S^{2n}$ has Hopf invariant 2.

Question: Hopf invariant one problem. For which n do we have a Hopf invariant one map $S^{4n-1} \rightarrow S^{2n}$? The answer is $n = 1, 2, 4$ and this is related to the existence of H -space structures on the big spheres. The solution is more interesting than the problem and uses Adams spectral sequence and the Steenrod algebra. Later, Atiyah proved the nonexistence of Hopf invariant one maps using K -theory and demonstrated the might of exotic cohomology theories.

Easy to show that $2n$ must be a power of 2 because it follows from Sq^i for $0 \leq i \leq p$ is 0 on $H^*(-; \mathbb{Z}/2)$ then the first nonzero Sq^j has j a power of 2. The Hopf invariant one map: define $X = e^0 \cup e^{2n} \cup e^{4n}$ with $(e^{2n})^2 = e^{4n}$.

9.3 Homotopy excision

Let X and Y be subcomplexes of a CW complex $Z = X \cup Y$, and let

$$A = X \cap Y.$$

Assume that the pairs (X, A) and (Y, A) are m -connected and n -connected respectively. Then inclusion of pairs induces a map

$$\pi_k(X, A) \longrightarrow \pi_k(X \cup Y, Y).$$

Theorem 9.3 (Homotopy excision). The map

$$\pi_k(X, A) \longrightarrow \pi_k(X \cup Y, Y)$$

is an isomorphism for $k < m + n$ and a surjection for $k = m + n$.

Remark 9.4. A useful special case is when $Y = CA$ is the cone on A . Then

$$X \cup_A CA \simeq X/A,$$

and since CA is contractible, we get

$$\pi_k(X \cup_A CA, CA) \cong \pi_k(X/A).$$

Thus homotopy excision gives a comparison map

$$\pi_k(X, A) \longrightarrow \pi_k(X/A)$$

which is an isomorphism for $k < m + n$ and a surjection for $k = m + n$.

Example 9.5. Suppose

$$X = A \cup_f e^{m+1}, \quad Y = A \cup_g e^{n+1}.$$

Then (X, A) is m -connected and (Y, A) is n -connected. In this case the theorem implies that

$$\pi_k(X, A) \oplus \pi_k(Y, A) \longrightarrow \pi_k(X \cup Y, A)$$

is an isomorphism for $k < m + n$ and a surjection for $k = m + n$.

In low degrees the relative homotopy of $X \cup Y$ is just the sum of the two pieces. Beyond this range, interaction terms can appear; the first such new classes are related to Whitehead products.

Theorem 9.6 (Freudenthal suspension theorem). Let X be an $(n - 1)$ -connected CW complex, with $n \geq 1$. Then the suspension map

$$\Sigma: \pi_k(X) \longrightarrow \pi_{k+1}(\Sigma X)$$

is an isomorphism for $k < 2n - 1$ and a surjection for $k = 2n - 1$.

Example 9.7. Take $X = S^2$. Since S^2 is 1-connected, we may take $n = 2$. The theorem then says that

$$\pi_2(S^2) \xrightarrow{\cong} \pi_3(S^3),$$

which is indeed the isomorphism

$$\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}.$$

It also says that the suspension map

$$\pi_3(S^2) \twoheadrightarrow \pi_4(S^3)$$

is surjective. In fact,

$$\pi_3(S^2) \cong \mathbb{Z}, \quad \pi_4(S^3) \cong \mathbb{Z}/2.$$

Idea of proof. Apply the homotopy excision theorem to the decomposition

$$\Sigma X = CX_+ \cup_X CX_-,$$

where CX_+ and CX_- are two copies of the cone on X , glued along their common base X .

Since each cone is contractible, the pair $(CX_{\pm}, X)$ is n -connected when X is $(n-1)$ -connected. Thus Blakers–Massey gives that the natural map

$$\pi_k(CX_+, X) \longrightarrow \pi_k(\Sigma X, CX_-)$$

is an isomorphism for $k < 2n$ and a surjection for $k = 2n$.

Now use the identifications

$$\pi_k(CX_+, X) \cong \pi_{k-1}(X), \quad \pi_k(\Sigma X, CX_-) \cong \pi_k(\Sigma X),$$

the second because CX_- is contractible. Under these identifications, the map above becomes exactly the suspension map

$$\pi_{k-1}(X) \longrightarrow \pi_k(\Sigma X).$$

□

9.4 Whitehead brackets

Example 9.8 (Whitehead products). Let us compute $\pi_3(\bigvee_{\alpha} S_{\alpha}^2)$, showing that it is free abelian with basis consisting of the Hopf maps $S^3 \rightarrow S_{\alpha}^2 \subset \bigvee_{\alpha} S_{\alpha}^2$ together with the attaching maps $S^3 \rightarrow S_{\alpha}^2 \vee S_{\beta}^2 \subset \bigvee_{\alpha} S_{\alpha}^2$ of the cells $e_{\alpha}^2 \times e_{\beta}^2$ in the products $S_{\alpha}^2 \times S_{\beta}^2$, for all unordered pairs $\alpha \neq \beta$.

Suppose first that there are only finitely many summands S_{α}^2 . For a finite product $\prod_{\alpha} X_{\alpha}$ of path-connected spaces, the map $\pi_n(\bigvee_{\alpha} X_{\alpha}) \longrightarrow \pi_n(\prod_{\alpha} X_{\alpha})$ induced by inclusion is surjective, since

$\pi_n(\prod_\alpha X_\alpha) \cong \bigoplus_\alpha \pi_n(X_\alpha)$ is generated by the subgroups $\pi_n(X_\alpha)$. Thus the long exact sequence of homotopy groups for the pair $(\prod_\alpha X_\alpha, \bigvee_\alpha X_\alpha)$ breaks up into short exact sequences

$$0 \longrightarrow \pi_{n+1}\left(\prod_\alpha X_\alpha, \bigvee_\alpha X_\alpha\right) \longrightarrow \pi_n\left(\bigvee_\alpha X_\alpha\right) \longrightarrow \pi_n\left(\prod_\alpha X_\alpha\right) \longrightarrow 0.$$

These short exact sequences split, since the inclusions $X_\alpha \hookrightarrow \bigvee_\alpha X_\alpha$ induce maps $\pi_n(X_\alpha) \rightarrow \pi_n(\bigvee_\alpha X_\alpha)$, and hence a splitting homomorphism $\bigoplus_\alpha \pi_n(X_\alpha) \rightarrow \pi_n(\bigvee_\alpha X_\alpha)$.

Taking $X_\alpha = S_\alpha^2$ and $n = 3$, we get an isomorphism

$$\pi_3\left(\bigvee_\alpha S_\alpha^2\right) \cong \pi_4\left(\prod_\alpha S_\alpha^2, \bigvee_\alpha S_\alpha^2\right) \oplus \left(\bigoplus_\alpha \pi_3(S_\alpha^2)\right).$$

The factor $\bigoplus_\alpha \pi_3(S_\alpha^2)$ is free with basis the Hopf maps $S^3 \rightarrow S_\alpha^2$. For the other factor, we have

$$\pi_4\left(\prod_\alpha S_\alpha^2, \bigvee_\alpha S_\alpha^2\right) \cong \pi_4\left(\prod_\alpha S_\alpha^2 / \bigvee_\alpha S_\alpha^2\right).$$

The quotient $\prod_\alpha S_\alpha^2 / \bigvee_\alpha S_\alpha^2$ has 5-skeleton a wedge of spheres $\bigvee_{\alpha \neq \beta} S_{\alpha\beta}^4$ for $\alpha \neq \beta$, so

$$\pi_4\left(\prod_\alpha S_\alpha^2 / \bigvee_\alpha S_\alpha^2\right) \cong \pi_4\left(\bigvee_{\alpha \neq \beta} S_{\alpha\beta}^4\right),$$

which is free with basis the inclusions $S_{\alpha\beta}^4 \hookrightarrow \bigvee_{\alpha \neq \beta} S_{\alpha\beta}^4$. Hence $\pi_4(\prod_\alpha S_\alpha^2, \bigvee_\alpha S_\alpha^2)$ is free with basis the characteristic maps of the 4-cells $e_\alpha^2 \times e_\beta^2$. Via the injection

$$\partial : \pi_4\left(\prod_\alpha S_\alpha^2, \bigvee_\alpha S_\alpha^2\right) \rightarrow \pi_3\left(\bigvee_\alpha S_\alpha^2\right),$$

this means that the attaching maps of the cells $e_\alpha^2 \times e_\beta^2$ form a basis for the summand $\text{Im } \partial \subset \pi_3(\bigvee_\alpha S_\alpha^2)$. This finishes the proof for the case of finitely many summands S_α^2 . The case of infinitely many summands follows immediately since any map $S^3 \rightarrow \bigvee_\alpha S_\alpha^2$ has compact image, hence lands in a finite subwedge, and similarly for any homotopy between such maps.

The maps $S^3 \rightarrow S_\alpha^2 \vee S_\beta^2$ in this example are expressible in terms of a product in homotopy groups called the **Whitehead product**, defined as follows.

Definition 9.9. Given basepoint-preserving maps $f : S^k \rightarrow X$ and $g : S^\ell \rightarrow X$, let $[f, g] : S^{k+\ell-1} \rightarrow X$ be the composite

$$S^{k+\ell-1} \longrightarrow S^k \vee S^\ell \xrightarrow{f \vee g} X,$$

where the first map is the attaching map of the $(k + \ell)$ -cell of $S^k \times S^\ell$ with its usual CW structure. Since homotopies of f or g give rise to homotopies of $[f, g]$, we obtain a well-defined product $\pi_k(X) \times \pi_\ell(X) \rightarrow \pi_{k+\ell-1}(X)$.

The Whitehead bracket is graded skew-symmetric for $k, \ell \geq 2$, i.e.

$$[f, g] = (-1)^{k\ell} [g, f].$$

When $\ell = 1$, the Whitehead bracket $[f, g]$ is the difference between the action of g on f and the trivial action. If $k = \ell = 1$, the Whitehead bracket is the commutator of f and g in $\pi_1(X)$.

Example 9.10.

10 Apr 9

Fibrations (see Mitchell Serre fibrations), definitions and basic properties, LES in homotopy, examples, Path fibration, homotopy fiber products, homotopy fibers, Serre fibration, topological groups and group like monoids, action of ΩB on the fibers, principal fibrations and Puppe sequence ("derived" LES)

10.1

Recall that $F \hookrightarrow E \rightarrow B$ is a fiber bundle if it is locally trivial over B .

Theorem 10.1. A fiber bundle over a paracompact base has the homotopy lifting property.

A related statement is that if B is paracompact and E extends to a fiber bundle over $B \times I$, then E^{ext} is isomorphic as a fiber bundle to $E \times I \rightarrow B \times I$. We proved this when F is a compact manifold. In particular if the bundle was trivial at time 0, then it is trivial at time 1 and also all times in between. This is precisely the notion of a fibration.

Definition 10.2. A map $F \hookrightarrow E \rightarrow B$ is a fibration if it has the homotopy lifting property, i.e. for any space X and any homotopy $H : X \times I \rightarrow B$ and any lift $H_0 : X \times \{0\} \rightarrow E$ of $H|_{X \times \{0\}}$, there exists a lift $H^{ext} : X \times I \rightarrow E$ of H extending H_0 .

Theorem 10.3. Let B be path connected. Pick $*$ $\in E$ a base point and F the fiber containing $*$. A fibration induces a long exact sequence in homotopy

$$\cdots \rightarrow \pi_k(F) \rightarrow \pi_k(E) \rightarrow \pi_k(B) \rightarrow \pi_{k-1}(F) \rightarrow \cdots \rightarrow \pi_1(E) \rightarrow \pi_1(B) \rightarrow \pi_0(F) \rightarrow \pi_0(E) \rightarrow \pi_0(B)$$

Need to say some things about what is happening at the end.

Proof. We need to define the map $\partial : \pi_k(B) \rightarrow \pi_{k-1}(F)$. Everything basically hinges on the fact that the pair $(I^k, \partial I^k \setminus \text{face})$ is homeomorphic to (I^k, face) . The homotopy lifting property is so cleverly chosen that everything works. $\square$

Serre adopted the definition to CW X .

Definition 10.4. $p : E \rightarrow B$ is a Serre fibration if it has the homotopy lifting property for cubes of all dimensions.

Theorem 10.5. A Serre fibration has the HLP for all CW pairs.

Proof. Lift the homotopy to all cells in $X \setminus A$. Run into the homotopy lifting problem for a cube by the same trick as before. $\square$

Theorem 10.6. Any fiber bundle is a Serre fibration.

Proof. Induct on the dimension of a cube. Given a cube, subdivide to trivialize bundle locally. Surely a product is a Serre fibration. The faces of the subdivision are smaller cubes. By inductive assumption, can extend the lifting to the smaller cubes moving from left to right. Can extend over interiors by the same trick. $\square$

10.2 Geometric examples

Example 10.7. $G \hookrightarrow EG \rightarrow BG$ is a fiber bundle whose total space is contractible. Therefore from the LES we see that

$$\pi_k(G) \cong \pi_{k+1}(BG)$$

For example

$$\pi_k(S^1) \cong \pi_{k+1}(\mathbb{CP}^\infty)$$

is isomorphic to $\mathbb{Z}$ for $k = 1$ and 0 for $k > 1$. Another example is $\pi_k(SU(n)) \cong \pi_{k+1}(\mathbb{HP}^\infty)$ which is 0 for $k < 3$, $\mathbb{Z}$ for $k = 3$ and $\mathbb{Z}/2$ for $k = 4$ (the same as $\pi_4(S^3)$).

Example 10.8. $S^1 \hookrightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$. We have the homotopy groups $\pi_k \mathbb{CP}^n$ is $\mathbb{Z}$ for $k = 2$ and $\mathbb{Z}$ for $k = 2n + 1$ and equal to $\pi_k S^{2n+1}$ after that.

Example 10.9. $U(n) \hookrightarrow U(n+1) \rightarrow S^{2n+1}$. If $k < 2n$ then $\pi_k U(n) \cong \pi_k U(n+1)$ so we get stabilization of homotopy groups of unitary groups. In particular $\pi_k U = \lim \pi_k U(n)$ is 0 for k even and $\mathbb{Z}$ for k odd. This is Bott periodicity.

10.3 The path fibration

Let X be a pointed space. Consider

$$\begin{aligned} P_*^X(X) &= \{\gamma : I \rightarrow X \mid \gamma(0) = *\} \\ P_*^A(X) &= \{\gamma : I \rightarrow X \mid \gamma(0) = *, \gamma(1) \in A\} \end{aligned}$$

Proposition 10.10. The endpoint evaluation map $P_*^A(X) \rightarrow A$ is a fibration maybe. It is surely a Serre fibration.

The path fibration $P_*^X(X) \rightarrow X$ is a fibration with total space contractible and the fiber over $*$ is the based loop space ΩX . Therefore we get $\pi_k(\Omega X) \cong \pi_{k+1}(X)$. Should remind you of $EG \rightarrow BG$.

Proposition 10.11. The long exact sequence for the path fibration $P_*^X(X) \rightarrow X$ induces isomorphisms $\pi_k(\Omega X) \cong \pi_{k+1}(X)$ for all k . These are the usual isomorphisms given by adjunction. The long exact sequence for the path fibration $P_*^A(X) \rightarrow A$ is the LES of the pair (X, A) .

10.4 Basic homotopy calculus with fibrations

Definition 10.12. If $E' \rightarrow B'$ is a fibration and $f : B \rightarrow B'$ is a map, we can form the pullback $E' \times_{B'} B$ and get a fibration $E' \times_{B'} B \rightarrow B$ called the pullback of E' along f .

If $E, E' \rightarrow B$ are fibrations, a fibration map $E \rightarrow E'$ is a map over B . A fiber homotopy equivalence $f : E \rightarrow E'$ is a fibration map which admits a fiber homotopy inverse, i.e. a fibration map $g : E' \rightarrow E$ such that $f \circ g$ and $g \circ f$ are fiber homotopic to the identity.

Proposition 10.13. If f is fiber preserving and a homotopy equivalence, then it is a fiber homotopy equivalence.

Two out of three lemma, also known as the 5-lemma in homotopy. It follows from the LES and the 5-lemma.

Lemma 10.14. Given a map f of fibrations $E' \rightarrow B'$ to $E \rightarrow B$ and B, B' are path connected, if any two of the three maps $E' \rightarrow E$, $B' \rightarrow B$ and the induced map on fibers are weak homotopy equivalences, then so is the third.

10.5 Homotopy fibers, homotopy fiber products, converting maps to fibrations

Given a map $f : X \rightarrow Y$, we can factor it as $X \rightarrow E \rightarrow Y$ with maps $g : X \rightarrow E$ and $p : E \rightarrow Y$ with g a homotopy equivalence and p a fibration. Recall the codual result that any map can be

factored as a cofibration followed by a homotopy equivalence, the intermediary being the mapping cylinder. In general there is Eckmann Hilton duality between fibrations and cofibrations, homotopy and homology.

How do we construct $E \rightarrow Y$? Consider free paths in Y

$$P_Y^Y(Y) = \{\gamma : I \rightarrow Y\}$$

comes with two evaluation maps. Now let

$$E = X \times_Y P_Y^Y(Y) \rightarrow Y$$

where we use ev_0 for the fiber product and ev_1 for the map to Y . E receives an obvious map from $X \times_Y Y$.

Definition 10.15. Let $X, Z \rightarrow Y$ be maps. The homotopy fiber product $X \times_Y^h Z$ is the space

$$X \times_Y P_Y^Y(Y) \times_Y Z$$

where we use ev_0 for the fiber product with X and ev_1 for the fiber product with Z .

It is characterized in the homotopy category by: a map $W \rightarrow X \times_Y^h Z$ up to homotopy is the same as a pair of maps $W \rightarrow X$ and $W \rightarrow Z$ such that the two compositions to Y are homotopic.

10.6 H -spaces, James space, and loops on a suspension

11 Apr 16

Homotopy replacement of fiber bundles (principal fibrations, Puppe sequence). Moore and Eilenberg–MacLane spaces. Principal fibration theorem for Eilenberg–MacLane fibrations. Postnikov towers, Moore–cell decomposition of a space, Eckmann–Hilton duality in H^* and π_* .

11.1 Singular complex and functorial CW approximation

Last time we considered the adjunction

$$\text{Sing} : \mathbf{Top} \rightleftarrows \mathbf{sSet} : |-|,$$

with Sing the right adjoint and geometric realization the left adjoint. Thus there are natural transformations $L \circ R \rightarrow \text{Id}$ and $\text{Id} \rightarrow R \circ L$. In particular, for every space X there is a natural map

$$| \text{Sing}(X) | \rightarrow X,$$

which gives a functorial CW approximation. Moreover, Sing and $|-|$ become inverse equivalences on the homotopy category.

11.2 Group-like monoids and the bar construction

Recall the slogan from last time: in homotopy theory, group-like monoids are as good as groups. An A_∞ -space is the homotopy-theoretic replacement for a strictly associative topological monoid, and one can build its bar construction via the geometric realization of the simplicial bar construction.

There is an algebraic analog. Suppose A is an associative algebra and M, N are right and left A -modules. If one wants to compute the tensor product correctly, in the derived sense, one resolves and forms a bar complex. A universal way to think about this is:

$$M \otimes_A N \simeq (M \otimes_k N) \otimes_{A \otimes A^{\text{op}}} A,$$

so one resolves A functorially as an $A \otimes A^{\text{op}}$ -module. This leads to the bar resolution

$$A \longleftarrow A^{\otimes 2} \longleftarrow A^{\otimes 3} \longleftarrow A^{\otimes 4} \longleftarrow \cdots,$$

where the face maps are given by multiplying adjacent factors, and the differential is the alternating sum of the face maps.

For example, if $M = N = k$ with augmentation $A \rightarrow k$, one gets the complex

$$k \longleftarrow A \longleftarrow A^{\otimes 2} \longleftarrow \cdots,$$

whose homology computes

$$\text{Tor}_*^A(k, k) \cong k \otimes_A^{\mathbf{L}} k.$$

Any two such resolutions are quasi-isomorphic.

Topologically, if M is a monoid, one considers

$$* \longleftarrow M \longleftarrow M^2 \longleftarrow M^3 \longleftarrow \cdots,$$

and fills in the simplices by homotopies. If M acts trivially on a point, replacing all copies of M by $*$ gives BM ; replacing only one component gives EM .

More generally, if M acts on the right on X and on the left on Y , the **homotopy quotient / two-sided bar construction** is

$$X \times_M^h Y.$$

modeled by the simplicial space

$$X \times Y \rightrightarrows X \times M \times Y \rightrightarrows X \times M^2 \times Y \rightrightarrows \cdots.$$

A homotopy-associative action is the same as filling in the simplices in this bar construction.

There is also the stricter simplicial replacement: a simplicial diagram

$$* \leftarrow M \leftarrow M^2 \leftarrow M^3 \leftarrow \dots$$

whose vertical maps to the corresponding powers are homotopy equivalences. Thus:

$$\begin{aligned} \{\text{topological groups}\} &\longleftrightarrow \{A_\infty\text{-group-like monoids}\}, \\ \{\text{topological actions}\} &\longleftrightarrow \{A_\infty\text{-actions}\}. \end{aligned}$$

Likewise, ordinary quotients are replaced by homotopy quotients:

$$(X \times Y)/G \quad \leadsto \quad X \times_G^h Y \simeq \text{Bar}(X, G, Y).$$

If the action is free and $X \times Y \rightarrow (X \times Y)/G$ is a fiber bundle, then the comparison map from the homotopy quotient to the ordinary quotient is a homotopy equivalence.

11.3 Principal fibrations and homotopy fibers

By this definition, any fibration

$$F \hookrightarrow E \rightarrow B$$

is associated to a principal homotopy-group fibration. For a path-connected base, one has the path-loop fibration

$$\Omega B \hookrightarrow P_b B \rightarrow B$$

and the loop space ΩB acts on the fiber F_b .

A principal fibration is a fibration equipped with a translation action by a loop space. In particular, since principal bundles are classified, one has the usual picture

$$\begin{array}{ccc} E & \longrightarrow & EG \\ \downarrow & & \downarrow \\ B & \longrightarrow & BG \end{array}$$

for a principal G -bundle.

If $B \rightarrow X$ is a map and $b \in B$ maps to the base point $x \in X$, then one gets a strict homomorphism

$$\Omega_b B \rightarrow \Omega_x X,$$

and hence an action of ΩX on the homotopy fiber. The total space E is the homotopy fiber of the classifying map $B \rightarrow X$.

Consequently, for any fibration $F \rightarrow E \rightarrow B$ one obtains a Puppe sequence

$$\Omega E \rightarrow \Omega B \rightarrow F \rightarrow E \rightarrow B,$$

and by continuing to the left and right one gets a sequence of spaces in which the composite of any two consecutive maps is null-homotopic, with a specified null-homotopy, and any three consecutive terms exhibit the leftmost one as the homotopy fiber of the right-hand map. Taking π_0 gives the long exact sequence in homotopy. Moreover, the original fibration is principal if and only if this sequence can be continued one more step to the right.

11.4 Eilenberg–MacLane spaces

Let π be a group, with π abelian if $n > 1$. A space $K(\pi, n)$ is a connected space with

$$\pi_i(K(\pi, n)) = \begin{cases} \pi & i = n, \\ 0 & i \neq n. \end{cases}$$

It is unique up to weak homotopy equivalence.

For $n \geq 2$, the Hurewicz theorem gives

$$H_n(K(\pi, n); \mathbb{Z}) \cong \pi, \quad H_i(K(\pi, n); \mathbb{Z}) = 0 \quad (i < n).$$

Hence the universal class in degree n gives

$$H^n(K(\pi, n); \pi) \cong \text{Hom}(\pi, \pi),$$

and in particular a distinguished class corresponding to the identity map.

The main representability statement is

$$[X, K(\pi, n)] \cong H^n(X; \pi).$$

For $n = 1$ and π abelian this agrees with the classification of principal π -bundles:

$$[X, B\pi] = \{\text{isomorphism classes of principal } \pi\text{-bundles on } X\} = H^1(X; \pi)$$

The Yoneda lemma implies that $K(\pi, n)$ is unique up to isomorphism in the homotopy category. A sketch with CW complexes is as follows: if $K(\pi, n)$ has no cells below dimension n , then any map $X \rightarrow K(\pi, n)$ factors through X/X_{n-1} , and one gets a homomorphism

$$C_n(X) \rightarrow \pi$$

which vanishes on boundaries exactly when the map extends.

Also note that if π is abelian, then $B\pi$ is an abelian topological group up to homotopy, and so is $\Omega B\pi \simeq \pi$ in the appropriate range.

11.5 Principal fibration theorem and Postnikov towers

Theorem 11.1 (Principal fibration theorem). Let B be simply connected, and let

$$K(\pi, n) \simeq F \hookrightarrow E \xrightarrow{p} B$$

be a fibration with $n > 1$. Then the monodromy action of $\pi_1(B)$ on $\pi_n(F) \cong \pi$ is trivial, and p is a principal $K(\pi, n)$ -fibration up to fiber homotopy equivalence. Consequently there is a classifying map

$$c_E : B \rightarrow BK(\pi, n) \simeq K(\pi, n+1)$$

such that E is the homotopy fiber of c_E . Hence fiber homotopy equivalence classes of such fibrations are in bijection with

$$[B, BK(\pi, n)] \cong [B, K(\pi, n+1)] \cong H^{n+1}(B; \pi).$$

The corresponding class

$$k(E) \in H^{n+1}(B; \pi)$$

is called the n th Postnikov-invariant of the fibration.

If B is not simply connected, then the monodromy action ρ of $\pi_1(B)$ on $\pi_n(F) = \pi$ may be nontrivial. In this case, one has the n th Postnikov invariant $k(E) \in H^{n+1}(B; \pi_\rho)$, where π_ρ is the local system on B associated to the representation ρ .

Remark 11.2. Suppose X is a connected CW complex and $\mathcal{L}$ corresponds to a $\pi_1(X)$ -module A is a local coefficient system. Then the universal cover $\tilde{X}$ has a cellular chain complex $C_*(\tilde{X})$ and this is naturally a module over $\mathbb{Z}[\pi_1(X)]$, because deck transformations act on $\tilde{X}$.

Then homology with local coefficients is

$$H_*(X; \mathcal{L}) = H_*(C_*(\tilde{X}) \otimes_{\mathbb{Z}[\pi_1(X)]} A)$$

and cohomology with local coefficients is

$$H^*(X; \mathcal{L}) = H^*(\text{Hom}_{\mathbb{Z}[\pi_1(X)]}(C_*(\tilde{X}), A)).$$

Theorem 11.3 (Postnikov tower). For a simply connected space X with homotopy groups $\pi_2, \pi_3, \dots$, there are spaces $X_{\leq n}$ and maps

$$X \rightarrow X_{\leq n}$$

such that

1. the induced map $\pi_k(X) \rightarrow \pi_k(X_{\leq n})$ is an isomorphism for $k \leq n$;
2. $\pi_k(X_{\leq n}) = 0$ for $k > n$;

3. there are fibrations

$$K(\pi_{n+1}(X), n+1) \rightarrow X_{\leq n+1} \rightarrow X_{\leq n},$$

classified by classes $k_{n+1} \in H^{n+2}(X_{\leq n}; \pi_{n+1}(X))$ the $(n+1)$ st Postnikov k -invariants.

4. The natural map $X \rightarrow \varprojlim_n X_{\leq n}$ is a weak homotopy equivalence.

The Postnikov tower is thus built from Eilenberg–MacLane fibers step by step, and is dual to building a space by attaching cells or, homologically, by Moore spaces.

11.6 Dold–Thom and Moore spaces

The Dold–Thom theorem says that the infinite symmetric product detects homology. For a based space X , define

$$\mathrm{SP}^\infty(X) = \varinjlim_m X^m / S_m.$$

$\mathrm{SP}^\infty(X)$ is a connected topological abelian group

Theorem 11.4 (Dold–Thom). For a based connected CW complex X ,

$$\pi_n(\mathrm{SP}^\infty(X)) \cong \tilde{H}_n(X; \mathbb{Z}) \quad (n \geq 1)$$

Since $\mathrm{SP}^\infty(X)$ is a connected topological abelian group, all Postnikov invariants vanish, and we have a homotopy equivalence

$$\mathrm{SP}^\infty(X) \simeq \prod_{n \geq 1} K(\tilde{H}_n(X; \mathbb{Z}), n)$$

In particular, topological abelian groups decompose as products of Eilenberg–MacLane spaces.

Example 11.5. A point of $\mathrm{SP}^m(\mathbb{C}P^1)$ is an unordered m -tuple of points on $\mathbb{C}P^1$, i.e. an effective divisor of degree m on $\mathbb{C}P^1$. Such a divisor corresponds to a homogeneous polynomial of degree m up to scale, namely the polynomial whose zeros are those points with multiplicity. So $\mathrm{SP}^m(\mathbb{C}P^1) \cong \mathbb{C}P^m$. Passing to the colimit over m gives

$$\mathbb{C}P^\infty \simeq \mathrm{SP}^\infty(\mathbb{C}P^1).$$

In particular, the group structure on $BS^1 \simeq \mathbb{C}P^\infty$ which is only defined up to homotopy can be made into a strict group structure by identifying it with $\mathrm{SP}^\infty(\mathbb{C}P^1)$.

Dually, a Moore space $M(\pi, n)$ is a space with one prescribed reduced homology group:

$$\tilde{H}_n(M(\pi, n)) \cong \pi, \quad \tilde{H}_i(M(\pi, n)) = 0 \text{ for } i \neq n.$$

The Postnikov tower is in this sense dual to a cell decomposition / Moore-space decomposition of a space.

12 Apr 21

Proof of principal fibration theorem, self maps of Eilenberg–MacLane spaces, construction of postnikov towers, postnikov factorization of a map, postnikov towers in families, obstruction theorem, functoriality of postnikov towers, examples, spectral sequences

12.1 Construction of Postnikov tower

Let X be simply connected. Then there is a tower of fibrations $X_i \rightarrow X_{i-1}$ with fiber $K(\pi_i(X), i)$, and X maps to all of them.

Theorem 12.1. This exists for every simply connected space X . It is homotopy functorial in the sense that a map $X \rightarrow Y$ induces a map between the towers. Moreover, $\pi_k(X) \rightarrow \pi_k(X_i)$ is an isomorphism for $k \leq i$. The map $X \rightarrow \varprojlim X_i$ is a weak homotopy equivalence because the fiber of $X_{\leq m} \rightarrow X_{\leq n}$ are n -connected, so no lifting problem for $\leq n$ -spheres.

The construction of $X_{\leq n}$ is by killing homotopy groups above n by attaching $n+2$ -cells and higher. We claim that attaching an $n+2$ -cell to X by $\phi : S^{n+1} \rightarrow X$ kills the class of $[\phi]$ in $\pi_{n+1}(X)$ and does not change $\pi_{\leq n}(X)$. It does change things higher up, but then you can inductively kill.

This claim follows from the excision theorem. If $X' = X \cup_{\phi} e^{n+2}$ and X is connected, then the long exact sequence of the pair (X', X) gives

$$\cdots \rightarrow \pi_{k+1}(X) \rightarrow \pi_{k+1}(X') \rightarrow \pi_{k+1}(X', X) \rightarrow \pi_k(X) \rightarrow \pi_k(X') \rightarrow \cdots$$

By the excision theorem

$$\pi_k(X', X) \cong \pi_k(X'/X) \cong \pi_k(S^{n+2})$$

for $k \leq n+1$. Note that X connected implies that (X', X) is $n+1$ -connected. Hence $\pi_k(X) \rightarrow \pi_k(X')$ is an isomorphism for $k \leq n$ and the map $\pi_{n+2}(X', X) \rightarrow \pi_{n+2}(S^{n+2})$ is surjective, i.e. the connecting map kills $[\phi] \in \pi_{n+1}(X)$.

12.2 Postnikov factorization of a map

Let $f : X \rightarrow Y$ be a map with X simply connected.

Theorem 12.2. For each n , there exists a space $P_n(f)$ and a diagram which commutes up to homotopy

$$X \rightarrow P_n(f) \rightarrow Y$$

so that $\pi_k(X) \rightarrow \pi_k(P_n(f))$ is an isomorphism for $k \leq n$ and $\pi_k(P_n(f)) \rightarrow \pi_k(Y)$ is an isomorphism for $k > n+1$, and

$$\pi_{n+1}(X) \twoheadrightarrow \pi_{n+1}(P_n(f)) \hookrightarrow \pi_{n+1}(Y)$$

If $Y = *$, then $P_n(f) = X_{\leq n}$ is the n -th Postnikov truncation of X . The construction first requires us to kill $\ker f_* : \pi_{n+1}(X) \rightarrow \pi_{n+1}(Y)$ and then kill $\pi_{>n+1}(X)$ by attaching cells. Then

$$\begin{array}{ccc} X & \rightarrow & Y \rightarrow Y_{\leq n+1} \\ & \downarrow & \\ & & X'_{\leq n+1} \end{array}$$

I claim that there exists a map $X'_{\leq n+1} \rightarrow Y_{\leq n+1}$ making the diagram commute up to homotopy. This is because $X'_{\leq n+1}$ is obtained from X by attaching cells of dimension $n+2$ and higher, the attaching maps are null-homotopic in $Y_{\leq n+1}$. Then $P_n(f)$ is the homotopy pullback of $X'_{\leq n+1} \rightarrow Y_{\leq n+1}$ and $Y \rightarrow Y_{\leq n+1}$.

12.3 Families version of Postnikov tower

Proposition 12.3. Let

$$F \longrightarrow E \xrightarrow{p} B$$

be a Serre fibration of connected CW complexes, and assume that B and F are simply connected. Then one can choose Postnikov towers for E and F ,

$$\cdots \longrightarrow E_{\leq n} \longrightarrow E_{\leq n-1} \longrightarrow \cdots \longrightarrow E_{\leq 1}, \quad \cdots \longrightarrow F_{\leq n} \longrightarrow F_{\leq n-1} \longrightarrow \cdots \longrightarrow F_{\leq 1},$$

together with maps

$$p_{\leq n} : E_{\leq n} \longrightarrow B$$

and maps

$$F_{\leq n} \longrightarrow E_{\leq n},$$

such that for every $n \geq 1$ the following hold:

1. The maps

$$E \longrightarrow E_{\leq n}, \quad F \longrightarrow F_{\leq n}$$

are the n -th Postnikov truncation maps.

2. The diagram

$$\begin{array}{ccc} E & \longrightarrow & E_{\leq n} \\ p \downarrow & & \downarrow p_{\leq n} \\ B & \xlongequal{\quad} & B \end{array}$$

commutes up to homotopy.

3. The homotopy fiber of

$$p_{\leq n} : E_{\leq n} \longrightarrow B$$

is $F_{\leq n}$.

4. These fit into maps of fibration sequences

$$\begin{array}{ccccc} F_{\leq n} & \longrightarrow & E_{\leq n} & \xrightarrow{p_{\leq n}} & B \\ \downarrow & & \downarrow & & \parallel \\ F_{\leq n-1} & \longrightarrow & E_{\leq n-1} & \xrightarrow{p_{\leq n-1}} & B, \end{array}$$

where the vertical maps are the Postnikov structure maps.

We can deduce this from the Postnikov factorization of a cleverly chosen map. In particular, just take the factorization $E \rightarrow P_n(p) \rightarrow B$. We are now ready to state the obstruction theorem, which is the reason we are able to do anything in homotopy theory at all.

12.4 Obstruction theorem

Suppose that we have a strictly commuting square of a cofibration $A \hookrightarrow X$ and a fibration $E \rightarrow B$:

$$\begin{array}{ccc} A & \longrightarrow & E \\ \downarrow & & \downarrow \\ X & \longrightarrow & B \end{array}$$

The question is does there exist a lift $X \rightarrow E$ making the diagram commute? Note that if we omit the cofibration/fibration condition, then we can only ask for commutation up to homotopy. If $B = *$ we are asking for an extension of a map $A \rightarrow E$ to X . If $A = *$, we are asking for a lift of a map $X \rightarrow B$ to E .

Theorem 12.4. Assume that one of the following holds.

1. X is obtained from A by attaching only $n + 1$ cells.
2. E is a principal $K(\pi, n)$ -fibration over B simply connected.

Then the square defines an obstruction class $\omega^{n+1} \in H^{n+1}(X, A; \pi_n(F))$ whose vanishing is necessary and sufficient for the existence of a lift $X \rightarrow E$ making the diagram commute.

Moreover, if $\omega^{n+1} = 0$ and X is obtained from A by attaching only $n + 1$ cells, then the set of lifts up to homotopy form a torsor under $H^{n+1}(X, A; \pi_{n+1}(F))$. If $\omega^{n+1} = 0$ and E is a principal $K(\pi, n)$ -fibration, then the set of lifts up to homotopy form a torsor under $H^n(X, A; \pi)$.

Remark that this criterion can be used iteratively. Check for a relative cell decomposition of X relative to A and go one dimension at a time, or check for the relative Postnikov tower of E relative to B and go one Postnikov stage at a time.

Corollary 12.5. If either $A \subset X$ or $E \rightarrow B$ is a weak homotopy equivalence, then the lift exists and is unique up to homotopy.

This is clear for fibrations since $\pi_* F = 0$. For cofibrations, while $H^*(X, A; \pi_n(F)) = 0$ an extra check is needed because we have to go one step at a time.

Example 12.6. Consider the path-space fibration

$$K(\pi, n) \simeq \Omega K(\pi, n+1) \longrightarrow PK(\pi, n+1) \longrightarrow K(\pi, n+1),$$

whose total space is contractible. Up to homotopy this is the universal principal $K(\pi, n)$ -fibration

$$K(\pi, n) \rightarrow * \rightarrow K(\pi, n+1).$$

Given a map $f : X \rightarrow K(\pi, n+1)$, a lift to $*$ exists if and only if f is nullhomotopic. Under the natural bijection

$$[X, K(\pi, n+1)] \cong H^{n+1}(X; \pi),$$

the homotopy class of f corresponds to the pullback $f^*(u)$ of the universal class

$$u \in H^{n+1}(K(\pi, n+1); \pi).$$

Hence the obstruction to lifting f is precisely $f^*(u)$.

Application is the functoriality of Postnikov towers. Assume X and Y are simply connected and $f : X \rightarrow Y$ is a map. We want to show that there is a map $X_{\leq n} \rightarrow Y_{\leq n}$. The point is that $X_{\leq n}$ is obtained from X by attaching cells of dimension $n+2$ and higher, and the attaching maps are null-homotopic in $Y_{\leq n}$, so we can extend the map $X \rightarrow Y_{\leq n}$ to $X_{\leq n}$.

This statement is analagous to the cellular approximation theorem, which says that if X is a CW complex and Y is a space, then any map $X \rightarrow Y$ is homotopic to a cellular map.

Proof of the obstruction theorem. Consider the case of extending across cells. A lifting on each cell exists if and only if $[f \circ \phi_\alpha] \in \pi_n(E)$ is 0 for each attaching map $\phi_\alpha : S^n \rightarrow X$, and f is the map $A \rightarrow E$. We know that $[p \circ f \circ \phi_\alpha] = 0$ as classes in $\pi_n(B)$ which means that $[f \circ \phi_\alpha]$ came from $\pi_n(F)$. We have defined the class in $H^{n+1}(X, A; \pi_n(F))$. This class cannot be killed unless it was 0 since there are no higher cells.

As for the case of the principal fibration, the fibration $E \rightarrow B$ is classified by a map $B \rightarrow K(\pi, n+1)$, and the composition $X \rightarrow B \rightarrow K(\pi, n+1)$ defines a class in (X, A) , and a nullhomotopy of the map to $K(\pi, n+1)$ corresponds precisely to a relative lifting to E . $\square$

13 Apr 23

Compute k invariant of a postnikov tower. Spectral sequences and examples. $BU(n)$, $EU(n)$, $K(\mathbb{Z}, 3)$, ΩS^2 , ΩS^3 . The homotopy groups of S^3 . Serre's theorem about separating torsion and $\mathbb{Q}$ parts. This says that localization of π_* agrees with the localization of H^* for coefficients $\mathbb{Z} \subset R \subset \mathbb{Q}$. Computation of $K(\mathbb{Q}, n)$ and rational homotopy theory.

13.1

Recall the Hopf fibration $S^1 \rightarrow S^3 \rightarrow S^2$ whose long exact sequence has isomorphisms $\pi_k S^3 = \pi_k S^2$ for $k \geq 3$. Moreover $\pi_2(S^3) = 0$. The Postnikov tower has $S^3 \rightarrow S^2 \rightarrow K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$ and living over $\mathbb{CP}^\infty$ has $S_{\leq 4}^2 = S_{\leq 4}^3$ with fiber $K(\pi_4 S^3, 4)$ towering over $S_{\leq 3}^2 = S_{\leq 3}^3$ with fiber $K(\mathbb{Z}, 3)$.

We need to find the k -invariant $k_4 \in H^4(\mathbb{CP}^\infty, \mathbb{Z})$.

Consider $SU(2) \rightarrow S^2 = SU(2)/S^1$ whose next terms in the Puppe sequence are $K(\mathbb{Z}, 2) = BU(1) \rightarrow BSU(2) \rightarrow \dots$

$$\begin{array}{ccccccc} S^1 & \rightarrow & S^3 & \rightarrow & S^2 & \rightarrow & BU(1) \rightarrow BSU(2) \\ S^1 & \rightarrow & K(\mathbb{Z}, 3) & \rightarrow & S_{\leq 3}^2 & \rightarrow & K(\mathbb{Z}, 2) \rightarrow K(\mathbb{Z}, 4) \end{array}$$

and the rows are Puppe sequences and there are vertical arrows which identify with the functoriality of Postnikov towers, in particular we are taking the third truncation of the first row to get the second row. Then c_2 of the standard bundle over $BSU(2)$ maps to $-c_1^2$ of the standard bundle over $BU(1)$, so $k^4 = -c_1^2$.

13.2 Spectral sequences and examples

The main example is the Leray spectral sequence. It computes the cohomology of the total space of a fibration $F \rightarrow E \rightarrow B$ from the cohomology of the base and the fiber. We will assume B is simply connected. The starting idea is to use a cell decomposition of B and over each cell, the fibration is fiber homotopy equivalent to a product.

Know how to build a complex by filtering by the skeleta B_n and $E_n = p^{-1}(B_n)$. We can keep increasing n by the long exact sequence

$$\dots \rightarrow H^*(E_n, E_{n-1}; \mathbb{Z}) \rightarrow H^*(E_n; \mathbb{Z}) \rightarrow H^*(E_{n-1}; \mathbb{Z}) \rightarrow \dots$$

and we actually know how to describe $H^*(E_n, E_{n-1}; \mathbb{Z})$. It is the product

$$\prod_{n+1 \text{ cells in } B} H^{*-(n+1)}(F; \mathbb{Z})$$

This is not an efficient method because one runs into trouble inductively. It also fails to exploit the algebraic ring structure on $H^*(E; \mathbb{Z})$ and the module structure of $H^*(E; \mathbb{Z})$ over $H^*(B; \mathbb{Z})$. The

better thing to do is to carry out an order by order computation of cohomology of a filtered complex. The analogy is to use Taylor series to compute solutions of ODEs. Each step is computing H^* of a complex.

Theorem 13.1 (Spectral sequence of a filtered complex). Consider a cohomologically graded complex

$$\dots \rightarrow C^{m-1} \rightarrow C^m \rightarrow C^{m+1} \rightarrow \dots$$

with a decreasing filtration

$$\dots \supset F^p C^* \supset F^{p+1} C^* \supset \dots$$

satisfying the Hausdorff condition $\bigcap_p F^p C^* = 0$ and completeness condition $C^* \cong \bigcup_p F^p C^*$. Then there exist a sequence of pages $E_r^{p,q}$ bigraded abelian groups with differentials $d_r : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$ so that $E_{r+1}^{p,q} \cong H^{p,q}(E_r, d_r)$ and d_r are the successive terms of δ in F . Moreover, $E_1^{p,q} \cong H^{p+q}(F^p C^* / F^{p+1} C^*)$.

We also have a short exact sequence

$$0 \rightarrow \lim^1_{p+q=n-1} \bigoplus E_\infty^{p,q} \rightarrow \text{Gr}^F H^*(C^\bullet) \rightarrow \bigoplus_{p+q=*} E_\infty^{p,q} \rightarrow 0$$

The spectral sequence converges if $\lim^1 \bigoplus_{p+q=n-1} E_\infty^{p,q} = 0$. A sufficient condition for convergence is for each $\bullet$, $F_n C^\bullet = 0$ for n sufficiently large. One really hopes that the $\lim^1$ term vanishes. If it doesn't you probably set the problem up wrong.

Example 13.2. Consider $k[z]$ filtered by $F^n k[z] = z^n k[z]$. Then $\lim^1 = k[[z]]/k[z]$.

An aside. Let $F_n \rightarrow F_{n-1} \rightarrow \dots \rightarrow$ be a system of maps of abelian groups. Then $\lim^0$ and $\lim^1$ are the cohomologies of

$$0 \rightarrow \lim^0 \rightarrow \prod F_n \rightrightarrows \prod F_n \rightarrow \lim^1 \rightarrow 0$$

where the two maps are the identity and the shift map. One has the Mittag Leffler condition which says that if for each n the image of $F_{n+k} \rightarrow F_n$ stabilizes for k sufficiently large, then $\lim^1 F_n = 0$. In particular, if the maps are surjective, then $\lim^1 = 0$.

Example 13.3. The Leray spectral sequence of a fibration $F \rightarrow E \rightarrow B$. Filter $C^\bullet E$ by the kernels of $C^\bullet E \rightarrow C^\bullet E_{n-1}$, that is

$$F^n C^\bullet E = \ker(C^\bullet E \rightarrow C^\bullet E_{n-1})$$

Then the filtered pieces fit into a short exact sequence

$$0 \rightarrow F^n C^\bullet E \rightarrow F^{n-1} C^\bullet E \rightarrow C^\bullet E_n / C^\bullet E_{n-1} \rightarrow 0$$

Thus $\text{Gr}^n C^\bullet E \cong C^\bullet E_n / C^\bullet E_{n-1}$. Then

$$H^*(\text{Gr}^p C^\bullet E) \cong \prod_{ncells \sin B} H^{*-n}(F; \mathbb{Z})$$

In particular

$$E_1^{p,q} \cong C^p(B; H^q(F; \mathbb{Z}))$$

Assume we don't need twisted coefficients. Then the differential d_1 is just the differential in $C^*(B)$. Thus we get the Leray spectral sequence with E_2 page

$$E_2^{p,q} \cong H^p(B; H^q(F; \mathbb{Z}))$$

and converging to $H^*(E; \mathbb{Z})$. This is equivalent to saying that $\text{gr } H^{p+q}(E) = E_\infty^{p,q}$. Note that for spaces $p, q \geq 0$ and so the filtration in cohomological degree n has $\leq n$ steps.

This is an example of a first quadrant spectral sequence.

Theorem 13.4. The Leray sequence is compatible with cup products and stable cohomology operations.

14 Examples

Consider the fibration $U(n-1) \rightarrow U(n) \rightarrow S^{2n-1}$. Assume inductively that $H^*(U(n-1); \mathbb{Z}) \cong \mathbb{Z}v_1, v_3, \dots, v_{2n-3}$ with $|v_i| = i$ with relations $v_i^2 = 0$ and skew commutativity. The spectral sequence is a grid with the v_i 's on the q -axis and the volume form on the p -axis. Looking at the numerics, there is nowhere for the differential to go, so all of the v_i 's survive to E_∞ .

So we see that

$$H^*(U(n); \mathbb{Z}) \cong H^*(U(n-1); \mathbb{Z})[v_{2n-1}]$$

which skew squares to 0. Here there is no ambiguity in lifting. Get free exterior algebra.

In this example, nothing interesting happens because of numerical reasons. Another example where nothing happens is $U(n)/U(n-1) \rightarrow BU(n-1) \rightarrow BU(n)$. If you know $BU(n-1)$ has cohomology $\mathbb{Z}[c_1, \dots, c_{n-1}]$ with $|c_i| = 2i$, then the spectral sequence has a nontrivial differential d_{2n-1} which kills the volume form on S^{2n-1} . In this case, Leray becomes the Gysin sequence of the sphere bundle $S^{2n-1} \rightarrow BU(n-1) \rightarrow BU(n)$.

Another example is $U(n) \rightarrow EU(n) \rightarrow BU(n)$. The claim is that the differentials take v_{2i-1} to $c_i \in H^{2i}$. This is an example of the Koszul complex which says that there is a complex $\Lambda^\bullet V \otimes S^\bullet V$ resolving k with differential taking the odd generators to the even generators.

15 Apr 28

Example 15.1 (The Hopf fibration). Consider the Hopf fibration

$$S^1 \longrightarrow S^{2n+1} \longrightarrow \mathbb{C}P^n.$$

This is a principal $K(\mathbb{Z}, 1)$ -fibration. The Leray–Serre spectral sequence has

$$E_2^{p,q} = H^p(\mathbb{C}P^n; H^q(S^1; \mathbb{Z})).$$

Writing

$$H^*(\mathbb{C}P^n; \mathbb{Z}) \cong \mathbb{Z}[\omega]/(\omega^{n+1}), \quad |\omega| = 2,$$

and writing θ for the generator of $H^1(S^1; \mathbb{Z})$, the E_2 -page is

$q = 1$	$\mathbb{Z}\theta$	0	$\mathbb{Z}\theta\omega$	0	$\cdots$	$\mathbb{Z}\theta\omega^n$
$q = 0$	$\mathbb{Z}$	0	$\mathbb{Z}\omega$	0	$\cdots$	$\mathbb{Z}\omega^n$
	$p = 0$	1	2	3	$\cdots$	$2n$

The only possible nonzero differential is $d_2: E_2^{p,1} \longrightarrow E_2^{p+2,0}$. Since the total space is S^{2n+1} , the classes in the bottom row except 1 must be killed. Thus

$$d_2(\theta) = \omega.$$

By multiplicativity,

$$d_2(\theta\omega^i) = \omega^{i+1}.$$

Thus the spectral sequence kills all intermediate cohomology and leaves exactly

$$H^0(S^{2n+1}; \mathbb{Z}) \cong \mathbb{Z}, \quad H^{2n+1}(S^{2n+1}; \mathbb{Z}) \cong \mathbb{Z}.$$

For a principal $K(\pi, n)$ -fibration $K(\pi, n) \rightarrow E \rightarrow X$ classified by a map $X \rightarrow K(\pi, n+1)$ the fundamental class $\iota \in H^n(K(\pi, n); \pi)$ transgresses to the corresponding characteristic class in $H^{n+1}(X; \pi)$. Equivalently, the Leray–Serre differential satisfies $d_{n+1}(\iota) = k$ where $k \in H^{n+1}(X; \pi)$ is the classifying k -invariant of the fibration.

Example 15.2 (The path fibration of $K(\mathbb{Z}, 3)$). Consider the path fibration

$$K(\mathbb{Z}, 2) \simeq \Omega K(\mathbb{Z}, 3) \longrightarrow PK(\mathbb{Z}, 3) \longrightarrow K(\mathbb{Z}, 3).$$

The total space is contractible. Let $\omega \in H^2(K(\mathbb{Z}, 2); \mathbb{Z})$ be the fundamental class of the fiber, and let $x \in H^3(K(\mathbb{Z}, 3); \mathbb{Z})$ be the fundamental class of the base. Since

$$H^*(K(\mathbb{Z}, 2); \mathbb{Z}) \cong \mathbb{Z}[\omega]$$

the E_2 -page begins as

$q = 6$	$\mathbb{Z}\omega^3$	0	0	$\mathbb{Z}x\omega^3$	0	0	$\dots$
$q = 5$	0	0	0	0	0	0	$\dots$
$q = 4$	$\mathbb{Z}\omega^2$	0	0	$\mathbb{Z}x\omega^2$	0	0	$\dots$
$q = 3$	0	0	0	0	0	0	$\dots$
$q = 2$	$\mathbb{Z}\omega$	0	0	$\mathbb{Z}x\omega$	0	0	$\dots$
$q = 1$	0	0	0	0	0	0	$\dots$
$q = 0$	$\mathbb{Z}$	0	0	$\mathbb{Z}x$	0	0	$\dots$
	$p = 0$	1	2	3	4	5	$\dots$

The first interesting differential is

$$d_3 : E_3^{p,q} \rightarrow E_3^{p+3,q-2}.$$

The transgression of the fundamental class is

$$d_3(\omega) = x.$$

By multiplicativity, d_3 is a derivation, so

$$d_3(\omega^m) = m\omega^{m-1}x.$$

In particular,

$$d_3(\omega^2) = 2\omega x, \quad d_3(\omega^3) = 3\omega^2 x.$$

Thus the d_3 -differential kills the multiples $2\omega x$, $3\omega^2 x$, $\dots$ and give rise to torsion residues on the next page, that is $\mathbb{Z}/2$ represented by ωx , a $\mathbb{Z}/3$ represented by $\omega^2 x$, and so on.

Since the total space $PK(\mathbb{Z}, 3)$ is contractible, none of these classes can survive to E_∞ . They must be killed by later differentials. This forces the base $K(\mathbb{Z}, 3)$ to have higher torsion cohomology classes.

$\pi_4(S^3)$ look at $S_{<3}^3 \rightarrow S^3 \rightarrow \text{somethings}$. There is a spectral sequence.

ΩX have a path fibration

$$\Omega X \rightarrow P_*^* X \rightarrow X$$

Caution if $\pi_1(X)$ is nontrivial, then it acts on $H^*(\Omega X)$ nontrivially by translation. In particular $H^0(\Omega X) = (\mathbb{Z}[\pi_1(X)])^\vee$ is the dual of the group ring of $\pi_1(X)$. So we can't invoke Leray yet. We will do a runaround. First let $X = S^2$.

$$\begin{array}{ccc} \eta^2/20 & & \\ u\eta 0 & & u\eta v \\ \eta 0 & & \eta v \\ u0 & & uv \\ \mathbb{Z}0 & & v \end{array}$$

Note that $u^2 = -u^2$ is 2 torsion so it cannot be the class that kills uv . So we conclude the existence of some η . We also have the calculation $d_2(u\eta) = v\eta - u d\eta = v\eta - uuv = v\eta$ so we conclude that $u\eta$ is the class living there. Also $\eta^2 = 2u\eta v$ but all the cohomology of the fibration must die so we conclude that $\eta^2/2$ is the class living there. In general

$$H^*(\Omega S^2) \cong \mathbb{Z}[\eta, \eta^2/2!, \eta^3/3!, \dots][u]$$

subject to $u^2 = 0$. Think of $\mathbb{Z}[u] = H^*(S^1)$. Remark that the $\Omega S^3 \rightarrow \Omega S^2 \rightarrow S^1 \rightarrow S^3 \rightarrow S^2 \rightarrow \mathbb{CP}^\infty$ and the claim is $\Omega S^3 \rightarrow \Omega S^2 \rightarrow S^1$ splits as a space. However, it does not split as a group because you could deloop and that would imply that $S^3 \rightarrow S^2$ splits as a space which is not true.

Homology of ΩS^2 . The ring structure from the Pontryagin product. ΩS^2 acts on the fibers of the path fibration. Then the spectral sequence gives us $\mathbb{Z}[U]$ with no divided powers, the generator coming from $\pi_1(\Omega S^2) = \pi_2(S^2)$. If you looked at S^3 , then the generator would be in even degree and so you wouldn't see any 2 torsion.

Definition 15.3. A connected space is abelian if $\pi_1(X)$ is abelian and acts trivially on $\pi_n(X)$ for $n \geq 2$.

Caution that S^2 is abelian by this definition but $[S^2, S^2]$ is not trivial, it is twice the generator of π_3 .

Definition 15.4. Let P be a set of primes, finite or not. An abelian group M is called P -primary torsion if $M[P^{-1}] = 0$.

Theorem 15.5. Let X be a connected abelian space. Then $\pi_*(X)$ is P -primary torsion if and only if $H_*(X; \mathbb{Z})$ is P -primary torsion. Equivalently, $\pi_*(X)[P^{-1}] = 0$ if and only if $H_*(X; \mathbb{Z}[P^{-1}]) = 0$.

Note that if P is all primes, then $\mathbb{Z}[P^{-1}] = \mathbb{Q}$. Taking all but 1 is the same as localizing at P .

Sullivan's arithmetic square says that if X is an abelian space, then there is a homotopy pullback square

$$\begin{array}{ccc} X & \rightarrow & X_{\mathbb{Q}} \\ & \Downarrow & \\ \prod \hat{X}_p & \rightarrow & \prod \hat{X}_{\mathbb{Q}_p} \end{array}$$

where $X_{\mathbb{Q}}$ is the rationalization of X and the hat means p completion. This relies on Serre's theorem to even make sense of the rationalization and to push a p -localization to a completion.

Proof. First we check that π_* implies H_* . induction on degree for all spaces. Hurewicz gives the base case. Look at the path fibration $\Omega X \rightarrow P_* X \rightarrow X$. Consider the spectral sequence with rows $H_{<n}(X)$ and columns $H_{<n-1}(\Omega X)$. Note that if $\pi_*(X)$ is P -primary torsion, then $\pi_*(\Omega X)$ is also

P -primary torsion because they have the same homotopy groups. Then by induction, $H_{<n-1}(\Omega X)$ is P -primary torsion which implies that $H_n(X)$ is P -primary torsion. The catch is that Leray may not apply because $\pi_1(X)$ may be nontrivial. We pass to the universal cover $\tilde{X}$ which has a free $\pi_1(X)$ action. The cochains on X are simply the $\pi_1(X)$ -invariants of the cochains on $\tilde{X}$. The claim is that if $\pi_1(X)$ is P -torsion, then $H_*(X, \mathbb{Z}[P^{-1}]) = H_*(\tilde{X}, \mathbb{Z}[P^{-1}])^{\pi_1(X)}$. This is simply an exercise in group cohomology.

Next we check the other direction. Proof for EM spaces by induction. Have the fibration $K(\pi, n) \rightarrow * \rightarrow K(\pi, n+1)$

stuff which is p primary torsion by assumption

$$\mathbb{Z}H_*K(\pi, n+1)$$

implies that $H_*K(\pi, n)$ is P -primary torsion. Iterate Leray, get the same thing for any truncation of Postnikov tower, inductively. Thus it is true for all H_* . $\square$

Have to be careful homology/cohomology of infinite CW complexes like $\mathbb{R}P^\infty$ like CW complexes. For example for $\mathbb{R}P^\infty$ the cohomology is simply $\mathbb{Z}/2$ in each degree but the K cohomology is

$$\mathbb{Z} \oplus_{\mathbb{Z}/2} \mathbb{Z}_2$$

in particular the inverse limit of torsion guys is not torsion.

Theorem 15.6. The rational homotopy of spheres is given by $\pi_*(S^{2n+1}) \otimes \mathbb{Q} = \mathbb{Q}$ in degree $2n+1$ and $\pi_*(S^{2n}) \otimes \mathbb{Q} = 0$ in degree $2n$ and $\mathbb{Q}$ in degree $4n-1$.

Use the path fibration of the spheres. $\Omega S^n \rightarrow * \rightarrow S^n$. For even spheres $\mathbb{Q}[x, y]$ with $\deg x = 2n$ and $\deg y = 4n-1$ and $dy = x^2$. Need a generator of $\pi_{2n-1}(\Omega S^{2n})$ odd to kill x another in $\pi_{4n-2}(\Omega S^{2n})$ to kill y .

Theorem 15.7. The rational cohomology of $K(\mathbb{Z}, n)$ is

$$H^*(K(\mathbb{Z}, n); \mathbb{Q}) \cong \mathbb{Q}[[v_n]]$$

where $v_n^2 = 0$ if n is odd.

Proof. Induction on n . Look at $K(\mathbb{Z}, n-1) \rightarrow * \rightarrow K(\mathbb{Z}, n)$. Get a Koszul sequence. If the base is a symmetric alg (exterior alg), fiber is an exterior alg (symmetric alg) there is no torsion left in differentials. $\square$

A good reference is Bott and Tu where he actually does exact couples. The point is that there are spectral sequences which are not from complexes, for example any spectral sequence involving homotopy groups.

16 April 30

Let P be a collection of primes in $\mathbb{Z}$. Let $R = \mathbb{Z}[P^{-1}]$ be the localization of $\mathbb{Z}$ at P . We said that a $\mathbb{Z}$ -module M is P -primary torsion if $M \otimes_{\mathbb{Z}} R = 0$.

Last time we did Serre's theorem which says that if X is an abelian space, then $\pi_*(X)$ is P -primary torsion if and only if $H_*(X; \mathbb{Z})$ is P -primary torsion. A complimentary result is the following.

Theorem 16.1. If $f : X \rightarrow Y$ is a map of abelian spaces then $f_* : \pi_*(X) \otimes_{\mathbb{Z}} R \rightarrow \pi_*(Y) \otimes_{\mathbb{Z}} R$ is an isomorphism if and only if $f_* : H_*(X; R) \rightarrow H_*(Y; R)$ is an isomorphism.

The group $H_*(X; R)$ can be identified with $H_*(X; \mathbb{Z}) \otimes_{\mathbb{Z}} R$ because R is flat over $\mathbb{Z}$, in particular $\text{Tor}_1^{\mathbb{Z}}(R, A) = 0$ for any abelian group A .

Proof. We show that π_* implies H_* . Let $F \rightarrow X \rightarrow Y$ be the homotopy fiber. The assumption on π_* implies that F has all P -primary torsion homotopy groups which implies that $H_*(F; \mathbb{Z}) = 0$ by Serre's theorem. The Leray sequeunce comparing $H_*(X; R)$ and $H_*(Y; R)$ implies that $H_*(F, \mathbb{Z}) \otimes_{\mathbb{Z}} R = 0$.

Caution that $\otimes$ commutes with all colimits but does not commute with inverse limits. In general

$$H^*(X; R) \not\cong H^*(X; \mathbb{Z}) \otimes_{\mathbb{Z}} R$$

Instead, in the special case when $R = \mathbb{Q}$, we get

$$H^*(X; \mathbb{Q}) \cong \text{Hom}_{\mathbb{Z}}(H_*(X; \mathbb{Z}), \mathbb{Q})$$

whence there is no Ext term here.

We show that H_* implies π_* . Take the same fibration $F \rightarrow X \rightarrow Y$. The Leray sequence in H_* wil show that $H_*(F)$ is all P -primary torsion. By Serre this would imply that π_F is all P -primary torsion. The LES implies that $\pi_*(X) \otimes_{\mathbb{Z}} R \rightarrow \pi_*(Y) \otimes_{\mathbb{Z}} R$ is an isomorphism.

Now an analysis of the E_2 page. All zero rows from $1, \dots, n-1$. If $H_n(F) \otimes_{\mathbb{Z}} R = 0$, then either there is a differential from $H_{n+1}(Y, R)$ to $H_n(F, R)$ (in which acse the map from $H_{n+1}(Y, R)$ to $H_{n+1}(X, R)$ is not an isomorphism) or there is no differential but then $H_n(F, R)$ survives to E_{∞} and gives a class in the kernel of $H_n(X, R) \rightarrow H_n(Y, R)$ which is not an isomorphism. Thus $H_*(F, R) = 0$. $\square$

Theorem 16.2. Let X be a CW complex, and let X^n denote its n -skeleton. Then for any coefficient group A ,

$$H_k(X; A) \cong \varinjlim_n H_k(X^n; A).$$

Moreover, there is a natural Milnor exact sequence

$$0 \rightarrow \varprojlim_n^1 H^{k-1}(X^n; A) \rightarrow H^k(X; A) \rightarrow \varprojlim_n H^k(X^n; A) \rightarrow 0.$$

For the skeletal filtration of a CW complex, the inverse system $H^k(X^n; A)\}_n$ is eventually constant, so $\varprojlim^1 = 0$. Hence

$$H^k(X; A) \cong \varprojlim_n H^k(X^n; A).$$

Proof. The key reason for stabilization is that the inclusion $X^n \hookrightarrow X$ induces

$$H^k(X; A) \xrightarrow{\cong} H^k(X^n; A)$$

whenever $n > k$. This is because the relative cohomology group $H^k(X, X^n; A)$ vanishes for $n > k$. This is because X/X^n has no cells in dimensions $\leq n$, so $H^k(X/X^n; A) = 0$ for $n > k$.

For fixed k , once the skeleton is past dimension k , the cohomology group no longer changes. Thus the inverse system is eventually constant, and the possible $\varprojlim^1$ obstruction disappears. $\square$

Example 16.3. Consider the infinite-dimensional complex projective space $\mathbb{CP}^\infty$. It has the CW structure with one cell in each even dimension. The cohomology groups are $\mathbb{Z}$ in even degrees and zero in odd degrees. This can be seen directly from the cellular chain complex, or from the skeleton filtration and the Milnor exact sequence.

Definition 16.4 (The R -local homotopy category of connected abelian spaces). Let X, Y be connected abelian spaces. Let $R = \mathbb{Z}[P^{-1}]$ be a localization of $\mathbb{Z}$. A map $f : X \rightarrow Y$ is an R -local homotopy if either of the following equivalent conditions hold:

- $f_* : \pi_*(X) \otimes_{\mathbb{Z}} R \rightarrow \pi_*(Y) \otimes_{\mathbb{Z}} R$ is an isomorphism.
- $f_* : H_*(X; R) \rightarrow H_*(Y; R)$ is an isomorphism.

Let $\text{Ho}(R)$ be the localization of the homotopy category at the class of R -local homotopy equivalences.

Note that for an $\mathbb{Z}$ -module M , being an R -module is a property, not a structure.

Theorem 16.5. There is a functorial R -localization of abelian spaces $X \mapsto X^R$ with the property that for any space Y with $\pi_*(Y)$ an R -module (equivalently for abelian spaces $H_*(Y; R)$ is an R -module) and map $f : X \rightarrow Y$, there is a unique map (up to homotopy) $X^R \rightarrow Y$ such that the obvious diagram commutes (up to homotopy).

The equivalence of the two conditions in the definition of R -local homotopy equivalence will quickly follow from the theorem because the homotopy fiber of $X \rightarrow X^R$ will have R -primary torsion homotopy groups and homology groups.

We give the construction of $X^{\mathbb{Q}}$ from rational cells. What is a rational cell? Well an ordinary cell is a cone on a sphere, so a rational cell is a cone on a rational sphere.

Definition 16.6. A rational n -cell for $n \geq 2$ is a cone over a rational $(n - 1)$ -sphere $S_{\mathbb{Q}}^{n-1}$.

Building $S_{\mathbb{Q}}^{n-1}$:

$$S^{n-1} \rightarrow S^{n-1} \rightarrow S^{n-1} \rightarrow \dots$$

take the colimit

$$\varinjlim S^{n-1}$$

with maps $k!$. The colimit is over integers k ordered by divisibility. This is the mapping telescope.

Proposition 16.7. $\tilde{H}_*(S_{\mathbb{Q}}^n; \mathbb{Z}) \cong \mathbb{Q}$ in degree n and 0 otherwise. If n odd, then $S_{\mathbb{Q}}^n \rightarrow K(\mathbb{Q}, n)$ induces an isomorphism on rational homology, which implies that $\pi_*(S_{\mathbb{Q}}^n) = H_*(S_{\mathbb{Q}}^n)$. If n is even we have a sequence

$$F \hookrightarrow S_{\mathbb{Q}}^n \rightarrow K(\mathbb{Q}, n) \rightarrow K(\mathbb{Q}, 2n)$$

where the last map is the square map. From the spectral sequence for H_* , H^* , we can see that $F = K(\mathbb{Q}, 2n - 1)$ principal fibration classified by a nonzero multiple of the square map.

Localizing a cell complex X with skeleta X_n . For abelian spaces, there is no distinction between based and free homotopy classes of maps $S^n \rightarrow X$. To rationalize X_1 we replace all circles with $S_{\mathbb{Q}}^1$. Will kill all torsion in π_1 next. Inductively,

$$\coprod S_{\mathbb{Q}, \alpha}^n \hookrightarrow \coprod D_{\mathbb{Q}, \alpha}^{n+1}$$

$$f_{\mathbb{Q}, \alpha} : \coprod S_{\mathbb{Q}, \alpha}^n \rightarrow X_{n, \mathbb{Q}}$$

$f_{\mathbb{Q}, \alpha}$ represent classes in $\pi_n(X_{n, \mathbb{Q}})$ which by inductive assumption $\pi_n(X_{n, \mathbb{Q}}) = \pi_n(X_n) \otimes_{\mathbb{Z}} \mathbb{Q}$. This implies that f_{α} extends to define unique maps $f_{\mathbb{Q}, \alpha} : S_{\mathbb{Q}, \alpha}^n \rightarrow X_{n, \mathbb{Q}}$ which we use to attach rational cells.

$X_{n+1}^{\mathbb{Q}}$ having rational homology implies that it has rational homotopy. The map $X_{n+1} \rightarrow X_{n+1}^{\mathbb{Q}}$ will be a $\mathbb{Q}$ -homology equivalence and therefore a $\mathbb{Q}$ -homotopy equivalence, so we can continue the next attachment.

Remark 16.8. If X is abelian, we can build a $\mathbb{Q}$ -cell model for $X^{\mathbb{Q}}$ with one cell for each basis element of $H_*(X; \mathbb{Q})$.

Remark 16.9. The construction for $X^{\mathbb{Q}}$ is not functorial as it is but there is a strictly functorial replacement.

16.1 Rationalization from the Postnikov tower

Recall the tower

$$X_{\leq 3} \rightarrow X_{\leq 2} \rightarrow X_{\leq 1} \rightarrow X_{\leq 0}$$

with homotopy fibers $K(\pi_n(X), n)$, classified by k -invariants $k^{n+1} \in H^{n+1}(X_{\leq n}; \pi_n(X))$. Assume all of the homotopy groups are finitely generated. Then $H^*(K(\pi_n(X) \otimes \mathbb{Q}, n); \mathbb{Q}) = H^*(K(\pi_n(X), n); \mathbb{Z}) \otimes \mathbb{Q}$ idunctively from the spectral sequence. Can proceed idncutvely. Rationalize all $|p_i^1$ up to n . The lery sequence means that inductively, we have rationalized up to $H^*(X_{\leq n}; \mathbb{Z})$. This implies that the rational k -invariant $k_{\mathbb{Q}}^{n+2} \in H^{n+2}(X_{\leq n}^{\mathbb{Q}}, \pi_{n+1}^{\mathbb{Q}})$ is uniquely defined, and then you can continue.

CAUTION! When computing the integral cohomology of rational EM spaces, the ansewr is not what you think it is. Work out $\text{Hom}(\mathbb{Q}, \mathbb{Z})$ and $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Q}, \mathbb{Z})$. This does not come up in the construction above.

Theorem 16.10 (Quillen, Sullivan). There is a triangle of equivalences of categories

$$\text{Ho}_{\text{simply connected}}(\mathbb{Q}) \cong \text{Ho}(\text{rational commutative dg algebras}) \cong \text{Ho}(\text{rational simply connected dg Lie algebras})$$

The map from (3) to (2) is given by the Chevalley–Eilenberg complex. The map from (1) to (2) is a version of the cochain complex. The map from (1) to (3) is a dgla version of L_{∞} -algebras, namely $(\pi_*^{\mathbb{Q}}, \text{whitehead})$.

Some remarks about DGLAs. A DGLA is a chain complex L with a Lie bracket $[-, -]$ satisfying the skew and graded Jacobi. It can be thought of as the Lie algebra of ΩX . The example to keep in the mind is the Whitehead bracket. The L_{∞} generalizatino allows higher brackets $b_n : L^{\otimes n} \rightarrow L$ satisfying some higher Jacobi identities. The relations allows one to define a differential on the DGCA $(\text{Sym}^{\bullet} L^{\vee}[1], d)$ which is the Chevalley–Eilenberg complex.

16.2 Two minimal models

Quillen minimal model. Let $L = \text{Free}(\mathbb{Q}\text{-cells in a minimal construction of } X^{\mathbb{Q}})$ where ∂ is the attaching map and ∂ sends new generators in dimension $n+1$ to h_* (free Lie algebra building the n -skeleton). This is a DGLA model for $X^{\mathbb{Q}}$. The differnetial has no linear part. The Sullivan minimal model $\mathbb{C}[\oplus \pi_k^{\mathbb{Q}}(X)]$ differential on π_k comes from the k -invariants of the attaching map $k^{n+2} \in H^{n+2}(X_{\leq n}^{\mathbb{Q}}, \pi_{n+1}^{\mathbb{Q}})$. and dualizes to a differential on $\mathbb{C}[\oplus \pi_k^{\mathbb{Q}}(X)]$.

Sullivan says that knowledge of a strictly commutative model for $C^*(X; \mathbb{Q})$ and the $\mathbb{Q}$ -lattice in $H^*(X; \mathbb{R})$ determines a rational model uniquely up to quasi-isomorphism. For example $(\Omega^*(X), d)$.